FINAL YEAR DESIGN PROJECT

Chase Your Need

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CHASE YOUR NEED(CYN)

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It is to certify that the final year design project (FYDP) of BSIT "Chase Your Need" was developed by **G7** under the supervision of "**Prof. Usman Ahmad**" in my opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in IT.

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Executive Summary

The Chase Your Need (CYN) mobile app is designed to bridge the gap between daily wage Service Providers. The app provides a simple, user-friendly platform where Service Providers, such as contractors and homeowners, can select the specific service according to their needs and then search for a nearby service provider that will fulfill their needs. Service providers, such as electricians, plumbers, carpenters, painters, welders, gardeners, and general service providers, can show their availability and get the work opportunity. The application allows Service providers to directly communicate with Service Providers through an integrated communication system, facilitating clear discussions about service requirements.

Developed using Flutter and Firebase as a backend, the app provides a smooth and user-friendly interface, supporting easy service. Workers / Service Providers get work opportunities by Clients, and communication between the two parties will be ensured. The application ensures a reliable and effective method for service providers to get work opportunities and clients to connect with skilled Service Providers quickly.

Acknowledgement

The acknowledgment section is an opportunity to express gratitude and appreciation to individuals and organizations who have contributed to the completion of your Final Year Project (FYP).

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Abbreviations

All the abbreviations and acronyms used in our FYP along with their respective brief description.

Abbreviation	Full Form	Explanation
CYN	Chase Your Need	The name of the mobile application that connects customers with nearby service providers.
UI	User Interface	The visual part of the application through which users interact with the system.
UX	User Experience	The overall experience and ease of use provided to the application users.
SDK	Software Development Kit	A collection of tools and libraries used to develop software applications.
API	Application Programming Interface	A set of rules that allows different software systems to communicate with each other.
DB	Database	A structured storage system used to save application data.
GPS	Global Positioning System	A location technology used to find the current position of customers and service providers.
FCM	Firebase Cloud Messaging	A Firebase service used to send push notifications to users.
OTP	One Time Password	A temporary code used for user verification and secure authentication.
APK	Android Package Kit	The file format used to install an Android application on a mobile device.
IDE	Integrated Development Environment	A software tool used by developers to write, test, and debug code.
JSON	JavaScript Object Notation	A lightweight data format used to store and exchange data between application and server.
REST API	Representational State Transfer API	A web service method used for communication between client and server systems.
HTTPS	Hypertext Transfer Protocol Secure	A secure communication protocol used to protect data transfer over the internet.
Auth	Authentication	The process of verifying a user's identity before allowing access to the system.
UID	User Identifier	A unique ID assigned to each registered user in the system.
URL	Uniform Resource Locator	The address used to access online resources such as images or files.
IOS	iPhone Operating System	Apple's mobile operating system for iPhone devices.

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Chapter 1: Introduction

1. Introduction

Chase Your Needs (CYN) is a mobile application developed to connect clients with nearby daily-wage service providers through a simple, location-based digital platform. In many areas, people still depend on personal contacts, neighborhood references, and informal hiring methods when they need urgent help for household, maintenance, transport, or manual work. This conventional approach is often slow, uncertain, and inconvenient because the required worker may be unavailable, difficult to locate, or unable to respond on time. CYN addresses this issue by providing one platform where clients can search for a required service, reach nearby available workers, communicate directly, and complete the hiring process in a structured manner.

The application is designed for two primary user roles: the client, who needs a service, and the service provider, who offers that service. After selecting a service category, the client can notify nearby providers, review their profiles, discuss the problem through in-app communication, and track the provider's arrival after the request is accepted.A major aim of the project is simplicity. The system is being designed with an interface that can be used by a wide range of people, including users with limited digital literacy.

1.1 Problem Statement

In Pakistan, many service providers work rely on traditional means of finding jobs, such as personal contacts, referrals, or waiting for customers to approach them. This creates uncertainty for workers and makes it difficult for clients to find the right person at the right time. In urgent situations, such as a plumbing fault, electrical issue, moving work, or mechanical problem, clients often waste their time calling familiar contacts who may not be available.

At the same time, many daily-wage workers face irregular employment because there is no simple and widely accessible platform dedicated to connecting them with nearby clients in real time. Existing systems either focus on formal jobs, support only limited-service areas, or present interfaces that are too complex for the target user group. As a result, there remains a need for a location-based and easy-to-use mobile application that can bridge the gap between clients and daily service providers.

1.2 Problem Solution

The proposed solution is the development of CYN as a location-based service marketplace application for mobile devices. The system allows users to register either as clients or as service providers. A client selects the required category of service, and the application identifies nearby providers based on their category, availability, and location. Relevant providers receive a service request notification and can accept or reject the opportunity within the application. Once a request is accepted, both parties are connected through in-app communication features so that the client can describe the problem and the service provider can clarify requirements, expected time, and approximate charges. After confirmation, the system shares real-time location information to help the client track the provider's arrival. When the work is completed, the client can mark the job as finished and submit a review and rating. This complete workflow transforms an informal and uncertain hiring process into a more transparent, efficient, and userfriendly digital system.

1.3 Objectives of the Proposed System

- To provide a simple and accessible platform that connects clients with nearby service providers.
- To reduce the time and effort required to find reliable workers for everyday service needs.
- To create more visible work opportunities for daily-wage service providers.
- To enable real-time communication between clients and service providers before delivery service.
- To support live location sharing so that both parties can track distance and expected arrival.
- To improve transparency through profile details, reviews, and ratings.
- To maintain a user-friendly system that can be used by people with different levels of digital literacy.
- To build a secure and scalable mobile application using modern development tools and backend services.

1.4 Scope

The scope of CYN covers the development of a cross-platform mobile application through which clients can request local services and service providers can receive work opportunities. The system includes user registration and login, profile management, service category browsing, provider listing, service request notifications, provider response handling, in-app chat and voice calling, real-time location sharing, and review submission after job completion.

The application is intended primarily for daily-life service needs, including but not limited to plumbing, electrician work, carpentry, painting, loading, driving, mechanical assistance, construction related labor, cleaning, and similar categories. Initially, the operational scope is expected to remain limited to selected cities or areas, with the possibility of future expansion to a wider geographic range.

1.5 System Components

1.5.1 AUTHENTICATION MODULE

This module manages user registration, login, logout, and password recovery. It ensures that each user can securely access the system according to the selected role of client or service provider.

1.5.2 PROFILE MANAGEMENT MODULE

This module stores and updates user information such as name, phone number, email address, location, service role, skills, experience, and availability. It allows users to maintain accurate profiles and helps clients evaluate providers before hiring.

1.5.3 SERVICE BROWSING AND MATCHING MODULE

This module enables clients to browse service categories, view nearby service providers, and inspect provider details such as service type, distance, experience, and reviews. It also supports the location-based matching process used to find relevant providers.

1.5.4 NOTIFICATION AND RESPONSE MODULE

This module delivers service requests to nearby providers and records whether the provider accepts or rejects the request. It acts as the main bridge between service demand and provider availability.

1.5.5 COMMUNICATION MODULE

This module supports direct interaction between clients and service providers through in-app chat and voice calling. Its purpose is to help both parties discuss the problem, negotiate the work, and coordinate service delivery.

1.5.6 LIVE LOCATION AND TRACKING MODULE

This module uses geolocation features to identify nearby service providers and later share real-time location data between both users after request acceptance. It improves convenience, transparency, and estimated arrival awareness.

1.5.7 REVIEW AND RATING MODULE

After completion of a service, this module allows the client to rate and review the service provider. These reviews support trust building and help future clients make more informed hiring decisions.

1.5.8 DATABASE AND BACKEND MODULE

This module manages persistent storage for user records, service details, service requests, communication logs, reviews, and related application data. The backend is planned around Firebase services to support authentication, cloud storage, and real-time synchronization.

1.6 Related System Analysis/Literature Review

A review of existing systems shows that several platforms already connect employers and workers, but many of them do not address the exact needs of daily-wage service providers. Platforms such as Rozee.pk primarily focus on formal employment and general job recruitment rather than immediate local services. Other applications, such as Karsaaz and Online Mazdoor, cover selected service categories or specific cities, but they may present limited accessibility, more complex interfaces, or restricted operating areas.

The analysis of these systems reveals an important gap. There is still a need for a simpler and more general application that supports nearby service discovery, real-time request handling, direct communication, and live location awareness in one easy workflow. CYN is proposed to address this gap by focusing specifically on everyday service access for clients and work availability for service providers.

1.7 Vision Statement

The vision of Chase Your Needs (CYN) is to provide a simple, reliable, and location-aware mobile platform that instantly connects clients with nearby skilled service providers, thereby improving access to daily services, increasing work opportunities for laborers, and promoting a more efficient and trustworthy service ecosystem.

1.8 System Limitations and Constraints

Like any practical software project, CYN operates under a number of limitations and constraints. At the initial stage, the system is expected to support only selected cities or limited geographic coverage. The platform depends heavily on internet connectivity for real-time features such as notifications, messaging, and live tracking. Geolocation accuracy may also vary depending on device capability and GPS conditions.

Another present limitation is the absence of in-app payment processing and formal skill verification. While profile details, ratings, and reviews can improve confidence, full professional verification mechanisms are outside the current project scope. In addition, user adoption may be gradual because many target users are accustomed to informal hiring methods. From a technical perspective, the application must remain lightweight, secure, scalable, and compatible across supported mobile devices.

1.9 Tools and Technologies

	Tools	Version	Rationale
Tools, Libraries, And Technologies	VS Code / Android Studio	Latest	Popular extensible IDEs for Flutter app development
	Libraries	**Version**	**Rationale**
	Flutter	Latest	Main framework for building cross-platform UI
	Provider	Latest	Simplifies the state management in Flutter
	http	Latest	Handles REST API class in Flutter
	cloud_firestore	Latest	Integrate Firestore database with Flutter
	firebase_auth	Latest	Handles Authentication using firebase in Flutter
	AgoraAPI	Latest	Handles voice and video call easily
	Technology	**Version**	**Rationale**
	Flutter	Latest	Enables modern, responsive, and scalable UI
	FireBase	Latest	Backend with real-time database, authentication, cloud functions

1.10 Project Deliverables

Here is a list of project deliverables for the Chase Your Need application:

1. Planning Document

- **Purpose:** Outlines the project goals, timelines, milestones, and resources required.
- **Contents:** Project scope, objectives, schedule, resource allocation, and risk management plan.
- **Appendix in FYP Report:** This document will be part of the initial planning phase and provide an overview of the project approach.

2. Software Requirements Specification (SRS)

- **Purpose:** Defines the functional and non-functional requirements of the system.
- **Contents:** Use cases, user stories, system features, system interfaces, and performance requirements.
- **Appendix in FYP Report:** This document will provide detailed specifications for the system, including all functional and non-functional requirements.

3. Prototypes

- **Purpose:** Visual representations of the user interface (UI) and user experience (UX).
- **Contents:** Interactive prototypes for the mobile application.
- **Appendix in FYP Report:** These prototypes will help visualize the user interface, which can be referenced for development and testing.

1.11 Project Planning

The project planning for CYN follows a phased approach beginning with requirement analysis and idea refinement, followed by interface design, system design, backend setup, feature implementation, integration, testing, and final documentation. Major functional modules include authentication, profile handling, service browsing, request notification, communication, live tracking, and review management.

The development effort is also distributed according to project needs such as frontend design, backend and database integration, communication module support, and documentation work. This phased planning helps ensure that the project remains organized and that each module is implemented and tested in a manageable sequence.

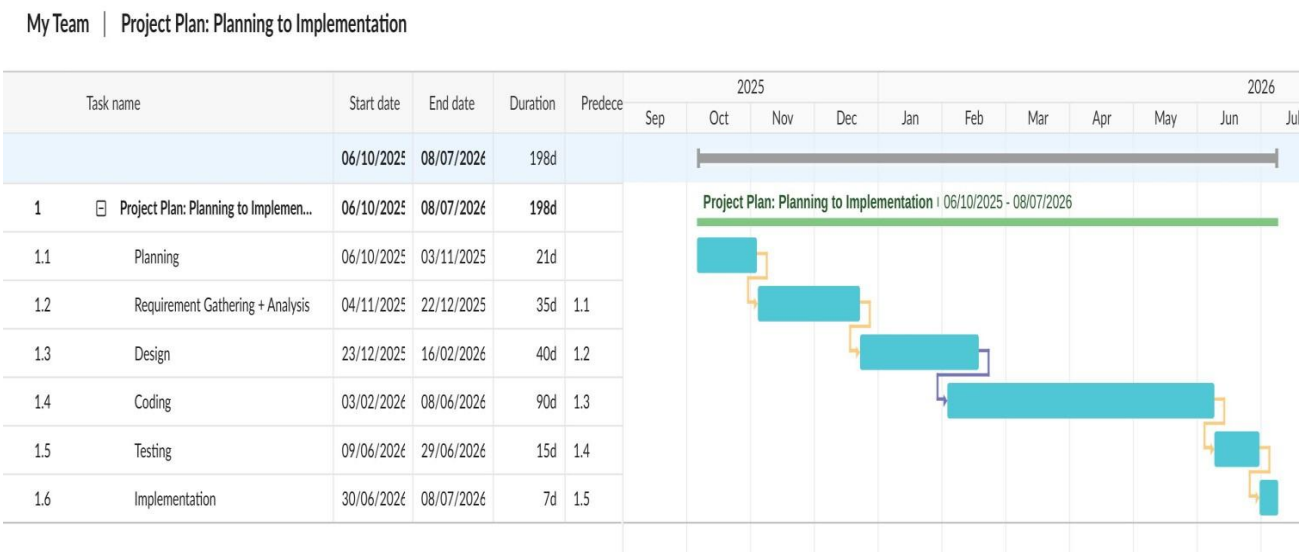


FIGURE 1 GANTT CHART

1.12 Summary

This chapter introduced the concept of Chase Your Needs (CYN) as a mobile application aim to connect clients with nearby daily-wage service providers through a simple and efficient digital workflow. The chapter described the problem addressed by the project, the proposed solution, system objectives, scope, major components, related systems, vision, limitations, technologies, deliverables, and planning direction.



Chapter 2: Requirements Analysis

2. Analysis

Chase Your Need (CYN) is a mobile application designed to connect customers with nearby local service providers. The main purpose of this application is to make it easier for users to hire workers such as electricians, carpenters, auto mechanics, masons, labour workers, loader rickshaw drivers, painters, plumbers, and other related service providers through a digital platform.

In the existing manual system, customers usually depend on personal contacts, recommendations, or visiting local markets to find workers. This process is time-consuming and does not always guarantee the availability or reliability of service providers. In emergency situations, such as electrical faults, plumbing problems, vehicle issues, or home repair needs, finding a nearby trusted worker becomes even more difficult.

The proposed CYN application provides a solution to this problem by allowing customers to search for nearby available service providers based on category and location. The system will identify the nearest available worker and send a service request to that provider. If the provider accepts the request, the customer and service provider will be able to communicate through chat, voice call, or video call. This communication will help the customer explain the problem clearly, and the service provider can discuss the required tools, material, estimated time, and service charges.

If the nearest provider rejects the request or does not respond, the system will forward the request to the next nearest available provider. This process will continue until an available provider accepts the request or no provider is available. After the service is completed, the customer can submit a rating and review. These ratings and reviews will help improve the service provider's profile and will help future customers make better decisions.

The application will be developed using Flutter for the mobile application interface and Firebase for backend services such as authentication, database management, real-time updates, notifications, and storage.

2.2 Requirement Identifying Technique

The requirements of the Chase Your Need application were identified using different requirement gathering techniques. These techniques helped in understanding the needs of customers, service providers, and the overall system.

OBSERVATION

Observation was used to understand how people currently search for local workers. It was observed that customers usually rely on phone contacts, personal references, or visiting local markets to find service providers. This method is slow, uncertain, and sometimes unreliable. It also becomes difficult when the customer needs urgent service.

INFORMAL INTERVIEWS

Informal discussions were conducted with potential customers and local service providers. Customers explained that they face difficulty in finding nearby and trusted workers. They also mentioned that it is difficult to know whether a worker is available or not. Service providers explained that they need a platform where they can receive more work opportunities and communicate easily with customers.

QUESTIONNAIRE

A questionnaire can be used to collect feedback from users regarding their expectations from the application. The questionnaire may include questions about required service categories, expected response time, importance of ratings, location-based searching, chat, calls, and service charges. This technique helps in identifying the features that users expect from the system.

EXISTING SYSTEM STUDY

Existing service booking and local worker hiring applications were studied to understand their features and limitations. This study helped in identifying useful features such as user registration, service category selection, provider profile, location tracking, request management, chat, and review system.

PROTOTYPE REVIEW

Initial user interface screens were designed and reviewed to identify the required modules of the application. The prototype included login, signup, provider dashboard, service request, provider profile, chat, live tracking, and review screens. Reviewing these screens helped in understanding the flow of the application and the required system features.

2.3 Functional Requirements

Functional requirements define the main features and services that the system must provide. These requirements describe what the application should do for customers and service providers.

The CYN application must allow users to register, log in, select service categories, search nearby providers, send service requests, accept or reject offers, communicate through chat or call, track provider location, complete jobs, and submit reviews.

2.3.1 Functional Requirement

Requirement ID	Functional Requirement	Description
FR-01	User Registration	The system shall allow customers and service providers to create accounts.
FR-02	User Login	The system shall allow registered users to log in using email/phone and password.
FR-03	Role Selection	The system shall allow users to select their role as Customer or Service Provider.
FR-04	Service Category Selection	The system shall allow customers to select service categories such as electrician, carpenter, auto mechanic, mason, labour, loader rickshaw, painter, and plumber.
FR-05	Location Access	The system shall access the user's current location to search nearby service providers.
FR-06	Search Nearby Providers	The system shall search available service providers based on selected category and nearest location.

FR-07	Send Service Request	The system shall send a service request to the nearest available service provider.
FR-08	Accept Request	The service provider shall be able to accept an incoming service request.
FR-09	Reject Request	The service provider shall be able to reject an incoming service request.
FR-10	Forward Request	If a service provider rejects the request, the system shall forward the request to the next nearest available provider.
FR-11	Chat Module	After request acceptance, the customer and service provider shall be able to communicate through chat.
FR-12	Voice and Video Call	The system shall provide voice and video call options between customer and service provider.
FR-13	Provider Tracking	The customer shall be able to view the provider's arrival status and estimated time.
FR-14	Job Completion	The system shall allow a service job to be marked as completed.
FR-15	Review and Rating	The customer shall be able to submit a rating and review after service completion.
FR-16	Provider Dashboard	The service provider shall be able to view job requests, completed jobs, ratings, and availability status.
FR-17	Profile Management	The system shall allow users to view and update their profile information.
FR-18	Notifications	The system shall send notifications for new requests, accepted requests, rejected requests, chat messages, and completed jobs.

2.4 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system. These requirements describe how the system should perform rather than what the system should do. The main non-functional requirements of the CYN application include reliability, usability, performance, and security.

2.4.1 Reliability

The CYN application should be reliable and should perform its functions correctly and consistently. The system must deliver service requests to the correct nearby available service provider. It should maintain accurate request statuses such as pending, accepted, rejected, in-progress, and completed.

The system should avoid duplicate service requests and should properly handle rejected requests by forwarding them to the next nearest available provider. Firebase real-time services will help keep data synchronized between customers and service providers. Chat messages, request status, provider availability, and job updates should be updated accurately in real time.

2.4.2 Usability

The application should be easy to use for both customers and service providers. The interface should be simple, clear, and user-friendly. Customers should be able to select a service category and send a service request with minimum steps.

Service providers should easily understand incoming service requests and respond by accepting or rejecting them. Buttons, icons, text fields, and navigation should be clear and understandable. Since many local workers may not be highly technical, the application should be designed in a simple way so that all users can operate it easily.

2.4.3 Performance

The CYN application should provide fast response time and smooth performance. The system should quickly search nearby available service providers based on the customer's selected category and location. Request delivery should be fast so that customers do not have to wait for a long time.

Chat messages and notifications should be delivered in real time. The application screens should load smoothly, and the system should handle multiple users and service requests without crashing. Locationbased searching should be optimized to provide quick and accurate results.

2.4.4 Security

The system should protect user data and ensure secure access. Firebase Authentication will be used for secure login and registration. User passwords should not be stored directly in the database.

The application should protect sensitive information such as phone numbers, email addresses, user locations, chat messages, and profile details. Database security rules should be applied so that users can only access authorized data. Only authenticated users should be able to create service requests, send messages, update profiles, and submit reviews.

2.5 External Interface Requirements

External interface requirements describe how the system interacts with users, software, hardware, and communication services. These requirements help define the connection between the CYN application and external components.

2.5.1 User Interfaces Requirements

The CYN application shall provide a clean and user-friendly mobile interface. The user interface should be designed for both customers and service providers. It should include readable text, clear buttons, simple navigation, proper icons, and attractive screen layouts.

The application shall include the following main screens:

Screen Name	Purpose
Login Screen	Allows users to log in to the application.
Create Account Screen	Allows new users to register as customer or service provider.
Customer Home Screen	Allows customers to select service categories.
Searching Screen	Shows that the system is searching for nearby providers.
Provider Profile Screen	Shows provider name, category, rating, experience, and contact options.
New Service Request Screen	Shows incoming service request details to the provider.
Chat Screen	Allows customer and provider to communicate through messages.
Tracking Screen	Shows provider location and estimated arrival time.
Job Completed Screen	Allows customer to submit review and rating.
Provider Dashboard	Shows provider jobs, rating, availability, and recent requests.
Profile Screen	Shows user personal information and settings.

2.5.2 Software Interfaces

The CYN application will use different software tools, frameworks, and services for development and backend functionality.

Software / Service	Purpose
Flutter	Used for developing the mobile application interface and logic.
Dart	Programming language used for Flutter development.
Firebase Authentication	Used for user login and registration.
Cloud Firestore / Firebase Realtime Database	Used for storing users, providers, service requests, chats, bookings, and reviews.
Firebase Cloud Messaging	Used for sending push notifications.
Firebase Storage	Used for storing profile images and chat images.
Google Maps API / Location Services	Used for location tracking and nearby provider searching.
Voice/Video Call SDK	Used for voice and video communication between customer and provider.

2.5.3 Hardware Interfaces

The CYN application will use mobile device hardware features to provide complete functionality.

Hardware Component	Purpose
GPS	Used to get customer and provider location.
Camera	Used to capture and send service issue images in chat.
Microphone	Used for voice and video calls.
Speaker	Used for call audio.
Touch Screen	Used for user interaction with the application.
Internet / Wi-Fi / Mobile Data	Used to connect with Firebase and communication services.

2.5.4 Communications Interfaces

The CYN application requires internet connectivity to communicate with Firebase backend services. The system will use secure communication channels to send and receive data such as login information, service requests, chat messages, provider location, and notifications.

The customer and service provider will communicate through chat, voice call, and video call after request acceptance. Firebase Cloud Messaging will be used to send notifications for new requests, accepted requests, rejected requests, chat messages, and job completion updates.

The application will also communicate with location services to get the current location of customers and service providers. This communication will help the system find nearby available providers and display estimated arrival information.

2.6 Summary

This chapter presented the requirements analysis of the Chase Your Need application. It explained the system analysis, requirement identification techniques, functional requirements, non-functional requirements, and external interface requirements.

The chapter identified the main features of the system, including user registration, login, role selection, service category selection, location-based provider search, request acceptance or rejection, request forwarding, chat, voice/video call, provider tracking, job completion, and review system.

It also discussed important quality requirements such as reliability, usability, performance, and security. The requirements defined in this chapter provide a clear foundation for the design and development of the CYN mobile application.



Chapter 3: Design and Architecture

3. System Design

The system design of Chase Your Need (CYN) describes the structure, components, data flow, user interaction, and overall architecture of the application. CYN is a mobile application that connects customers with nearby service providers such as electricians, carpenters, auto mechanics, masons, labour workers, loader rickshaw drivers, painters, plumbers, and other related workers.

The application is designed for two main types of users: customers and service providers. Customers can search for nearby workers according to the required service category and current location. Service providers can receive service requests, accept or reject them, communicate with customers, and complete jobs.

The system uses Flutter for mobile application development and Firebase for backend services. Firebase provides authentication, real-time database services, cloud storage, notifications, and user data management. The location-based service provider search allows the system to find the nearest available worker and send a request. If the worker accepts the request, a chat session is opened between customer and provider. If the worker rejects the request, the request is forwarded to the next nearest available provider.

The main goal of the system design is to create a reliable, user-friendly, real-time, and secure mobile application for local service hiring.

3.1 Design Considerations

The following design considerations were considered while designing the CYN application:

1. USER-FRIENDLY INTERFACE

The application should be simple and easy to use for both customers and service providers. Many local workers may not be highly technical, so the interface must be clear, simple, and understandable.

2. LOCATION ACCURACY

The system depends on the user's current location to find nearby service providers. Therefore, accurate GPS location is important for searching, tracking, and estimated arrival time.

3. REAL-TIME COMMUNICATION

The application requires real-time updates for service requests, chat messages, provider availability, and job status. Firebase real-time services will help provide quick synchronization between customer and provider.

4. AVAILABILITY MANAGEMENT

Service providers should be able to set their availability status. Requests should only be sent to providers who are currently available.

5. REQUEST FORWARDING

If the nearest service provider rejects the request or does not respond within a specific time, the system should forward the request to the next nearest available provider.

6. SECURITY AND PRIVACY

The system should protect user data such as name, phone number, email, location, chat messages, profile details, and service history. Only authenticated users should be allowed to access system features.

7. SCALABILITY

The system should be designed in a way that it can support multiple users, service categories, service providers, requests, and chat sessions in the future.

8. PERFORMANCE

The application should load quickly, search nearby providers efficiently, and deliver notifications and chat messages in real time.

3.2 Design Models

Design models are used to represent the structure and behavior of the system. These models help in understanding how different parts of the system interact with each other. The major design models of the CYN application include:

- Class Diagram
- Interaction Diagram
- State Transition Diagram
- UML Component Diagram
- Use Case Diagram
- MVC Model
- ER Diagram
- Behavioral Model

3.2.1 Class Diagram

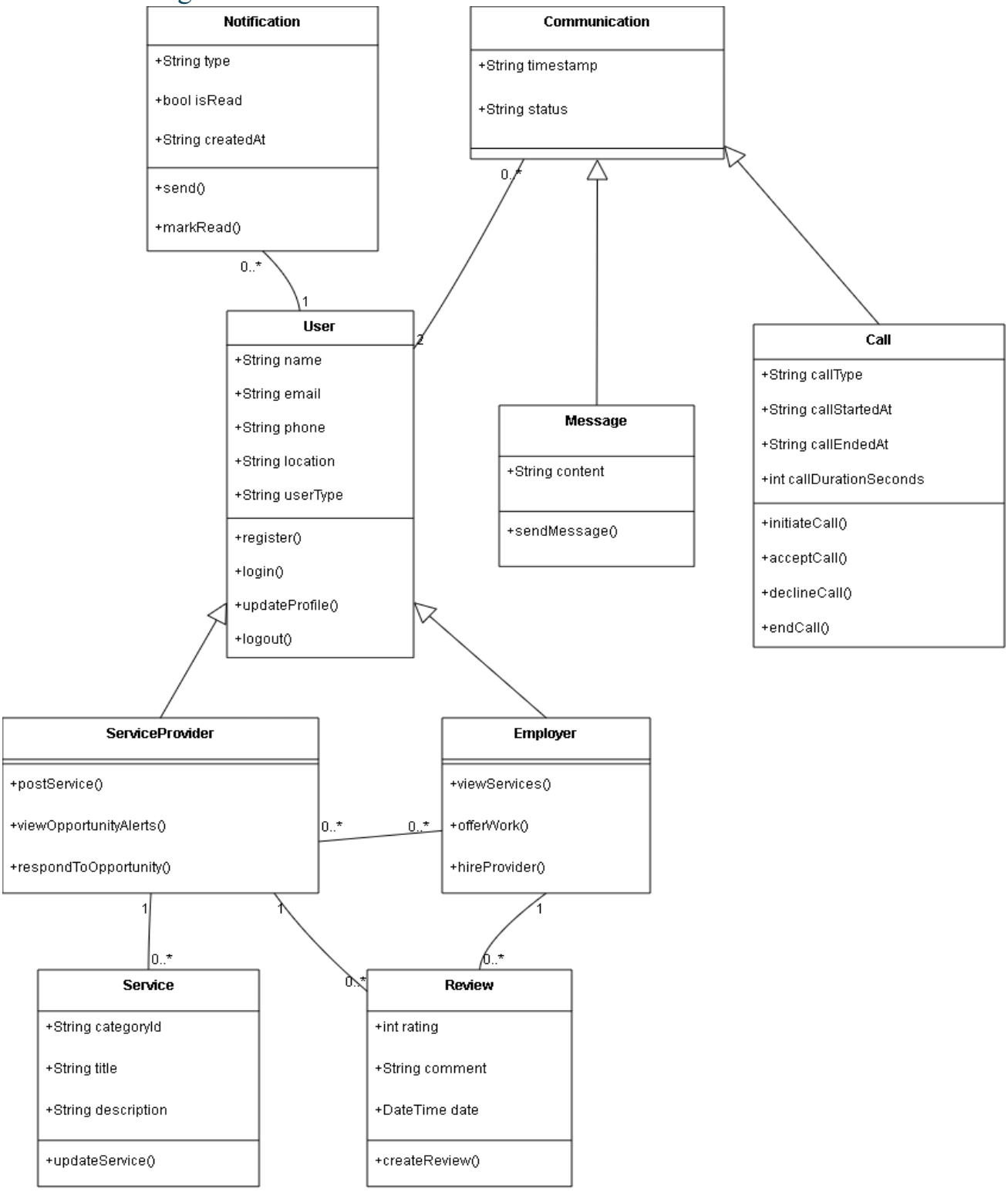


FIGURE 2 CLASS DIAGRAM

Class Diagram Explanation:

The class diagram of the CYN app shows the dependencies or relationships of classes.

1. **User:** This is the main class which includes the properties of userId, password, email, contact and related information.
2. **Service Provider**: Inherits from user, post his service details i.e. his expertise, view opportunity alerts and respond.
3. **Employer:** Inherits from user class, view services, offer-work and hire service provider.
4. **Service:** This class includes the service details of the worker expertise such as electrician, plumber, carpenter, auto-mechanics etc.
5. **Notification:** Detail of employer and to whom the notification is sent, job approval or rejection detail.
6. **Communication:** It includes the user data of communication messages and call related details.
7. **Review:** Employers can review the service provider. This is to increase the ratings of the workers.

Relationships:

User is the parent class of employer and the service provider which holds the data of both users. The service provider is attached to the service class, which is the details of his job expertise so that he would be added to the category of his work. The employer is attached to a review class so that he can review the worker about his service satisfaction. Both users are connected through a communication class to explain the kind of work so that the worker can bring all the necessary tools and equipment for work to be done. Notification class is for both users in which the employer will notify the worker, and the worker will get the notification of job approval or rejection.

3.2.2 Interaction Diagram

DIAGRAM 1: AUTHENTICATION (REGISTER & LOGIN)

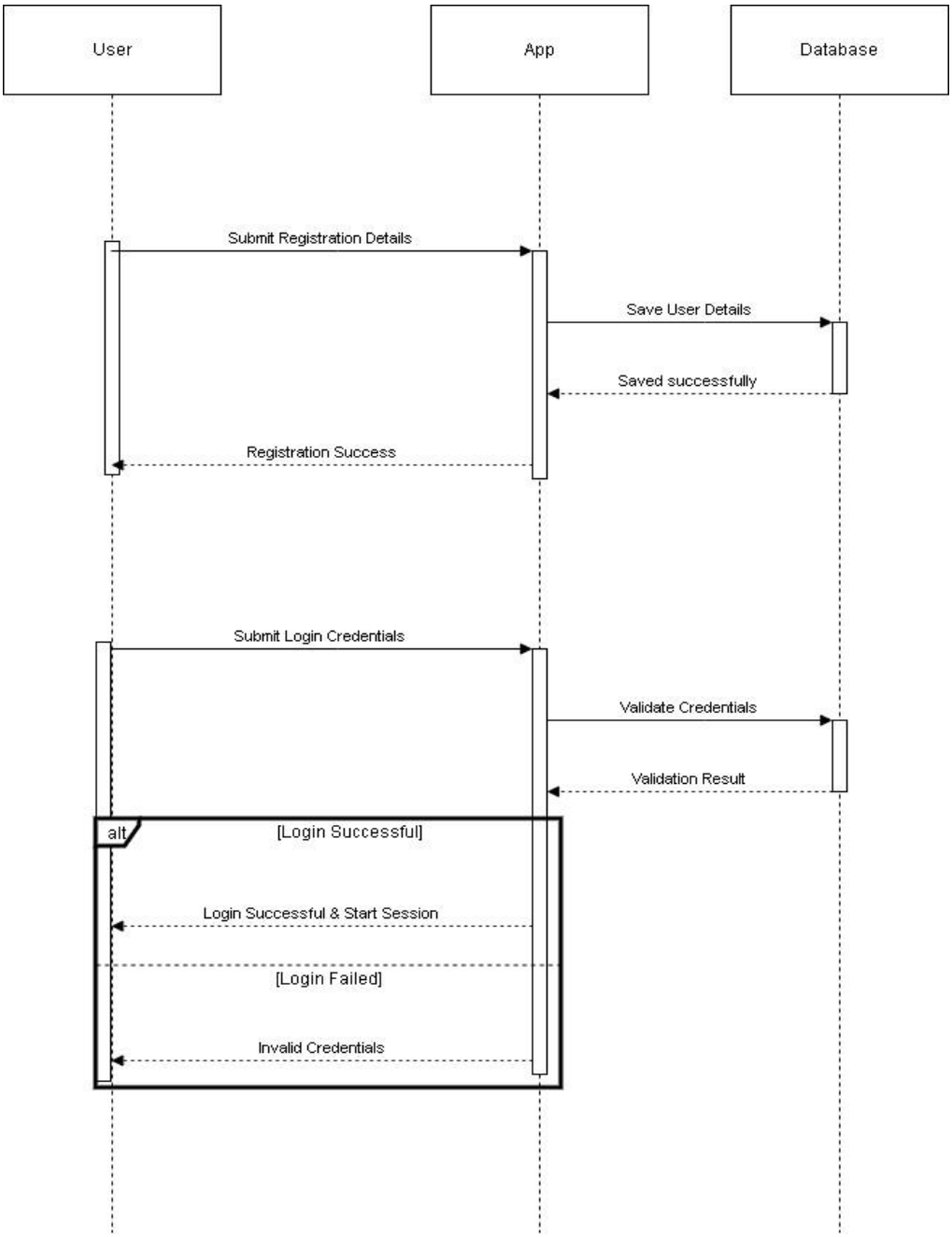


FIGURE 3 AUTHENTICATION INTERACTIONS

DIAGRAM 2: PROFILE MANAGEMENT

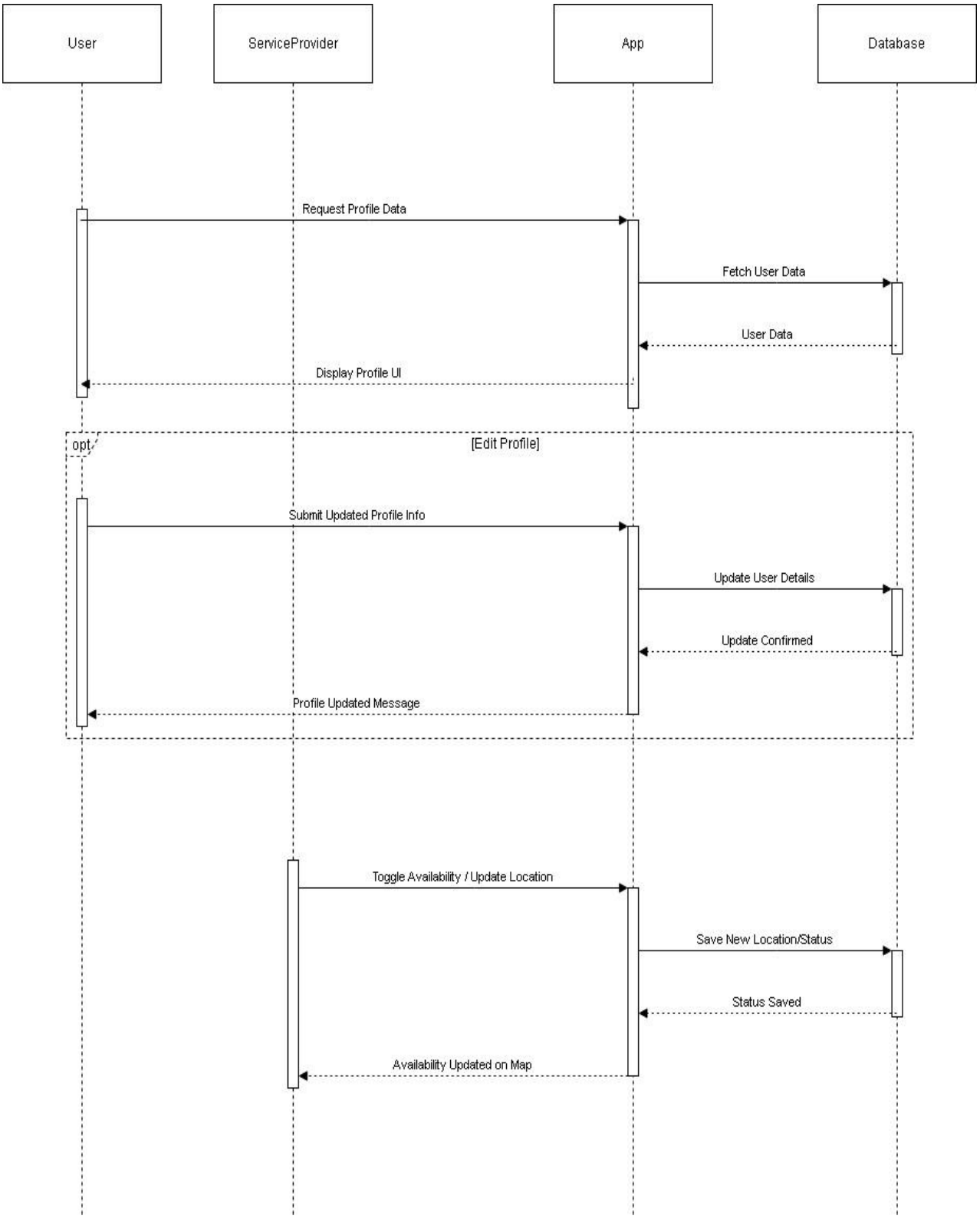


FIGURE 4 PROFILE MANAGEMENT

DIAGRAM 3: SERVICE MANAGEMENT

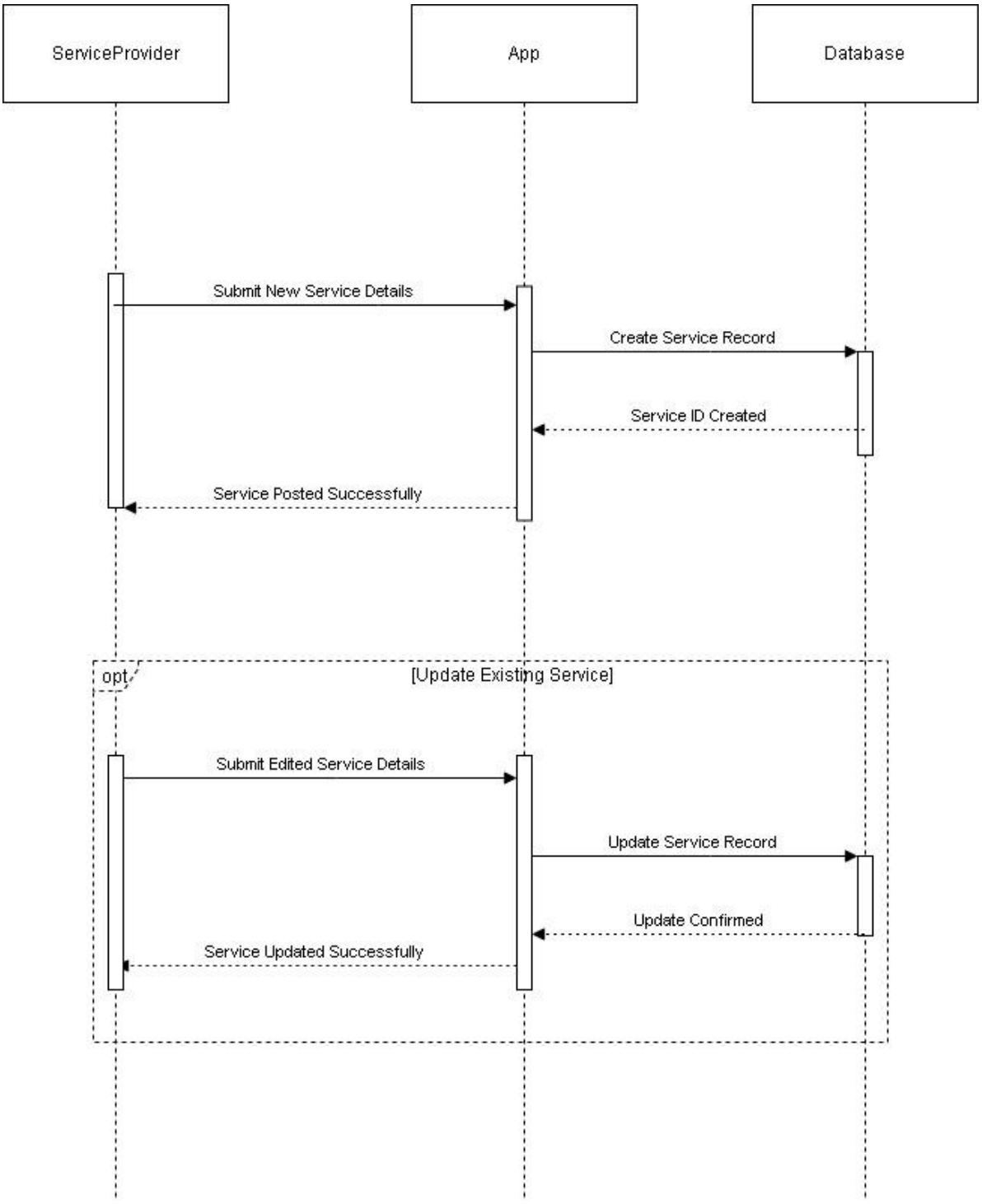


FIGURE 5 SERVICE MANAGEMENT

DIAGRAM 4: BROWSING SERVICES (BY EMPLOYER)

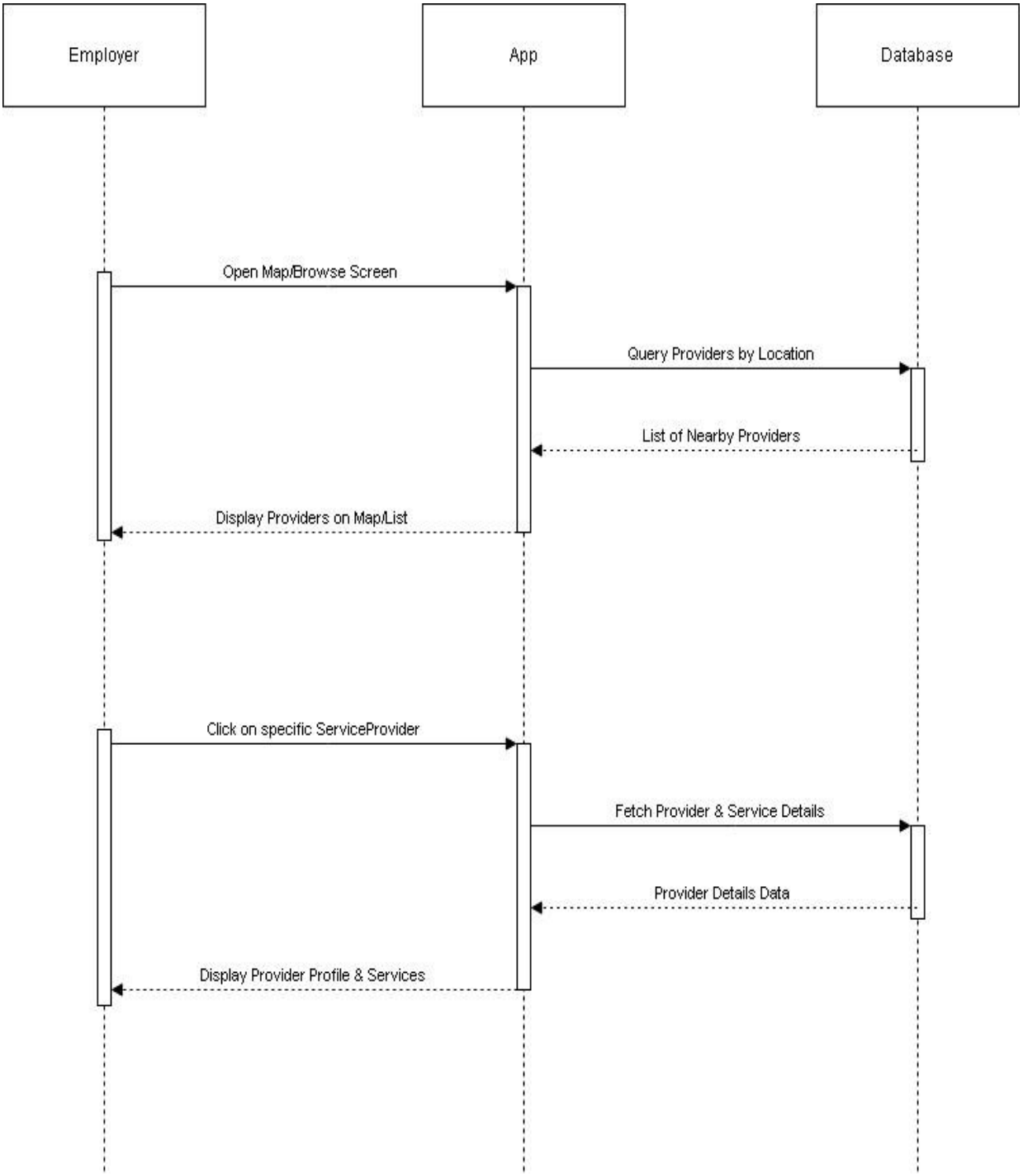


FIGURE 6 BROWSING SERVICES

DIAGRAM 5: COMMUNICATION

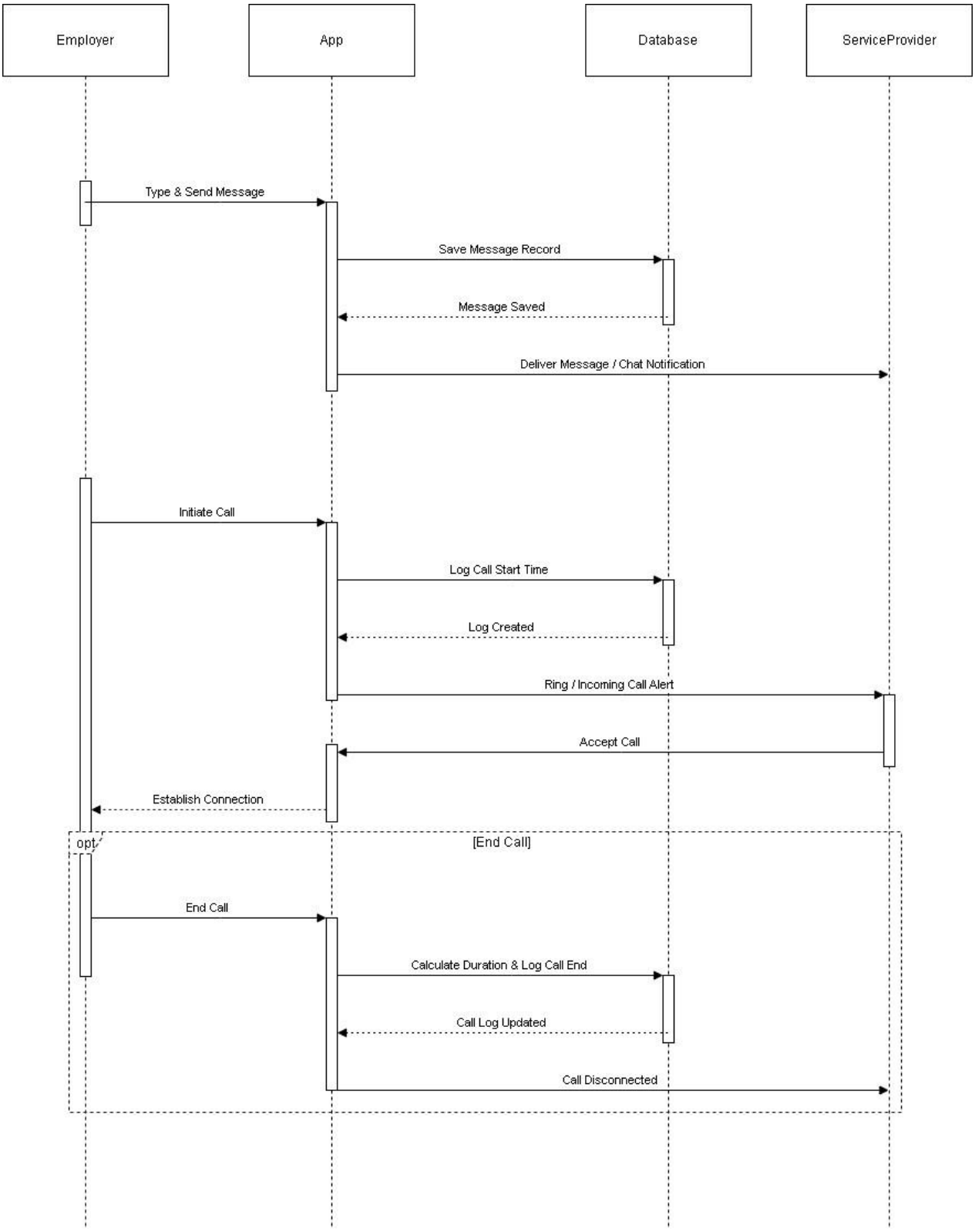


FIGURE 7 COMMUNICATION

DIAGRAM 6: JOB CLOSING

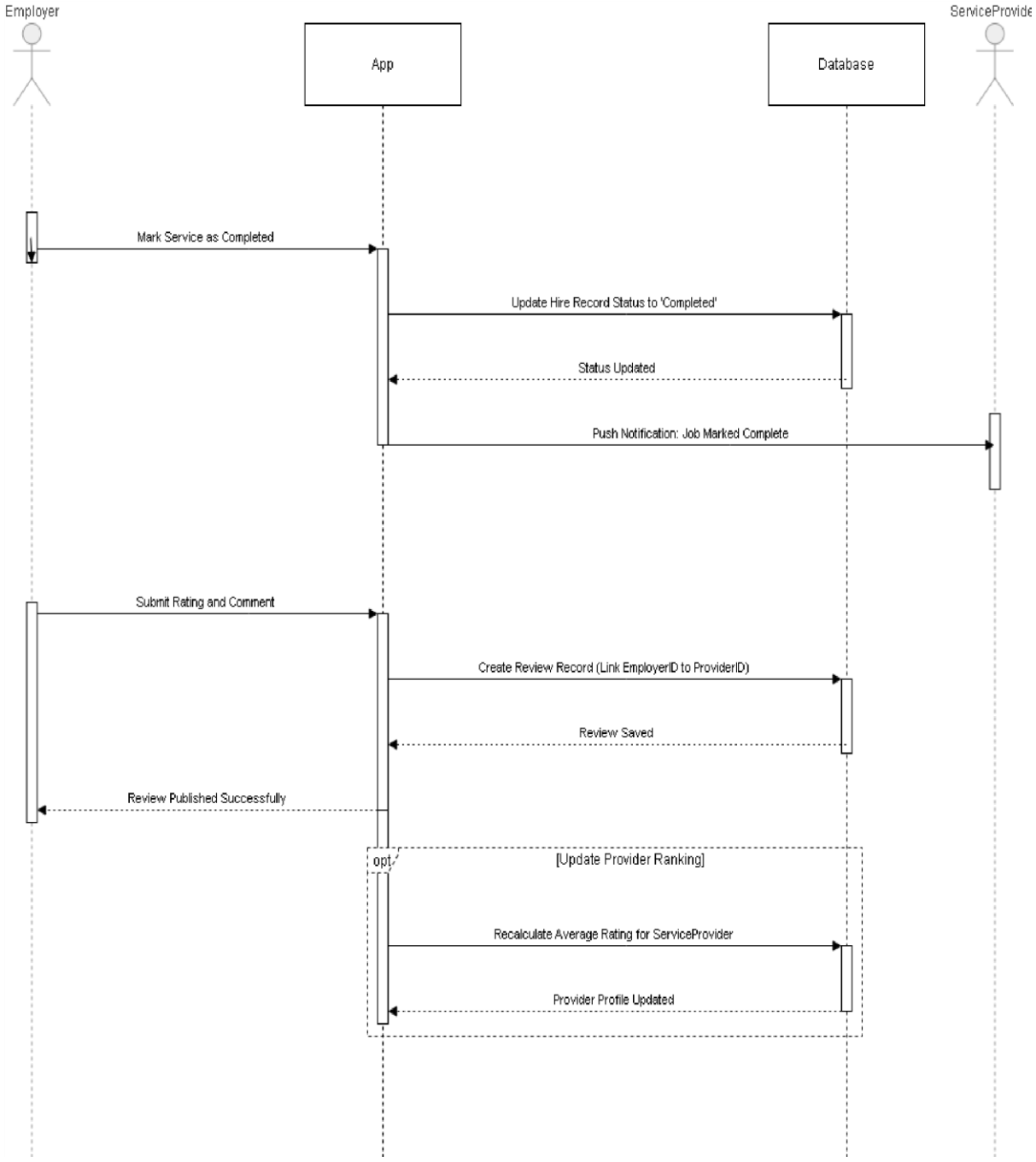


FIGURE 8 JOB CLOSING

3.2.3 State Transition Diagram

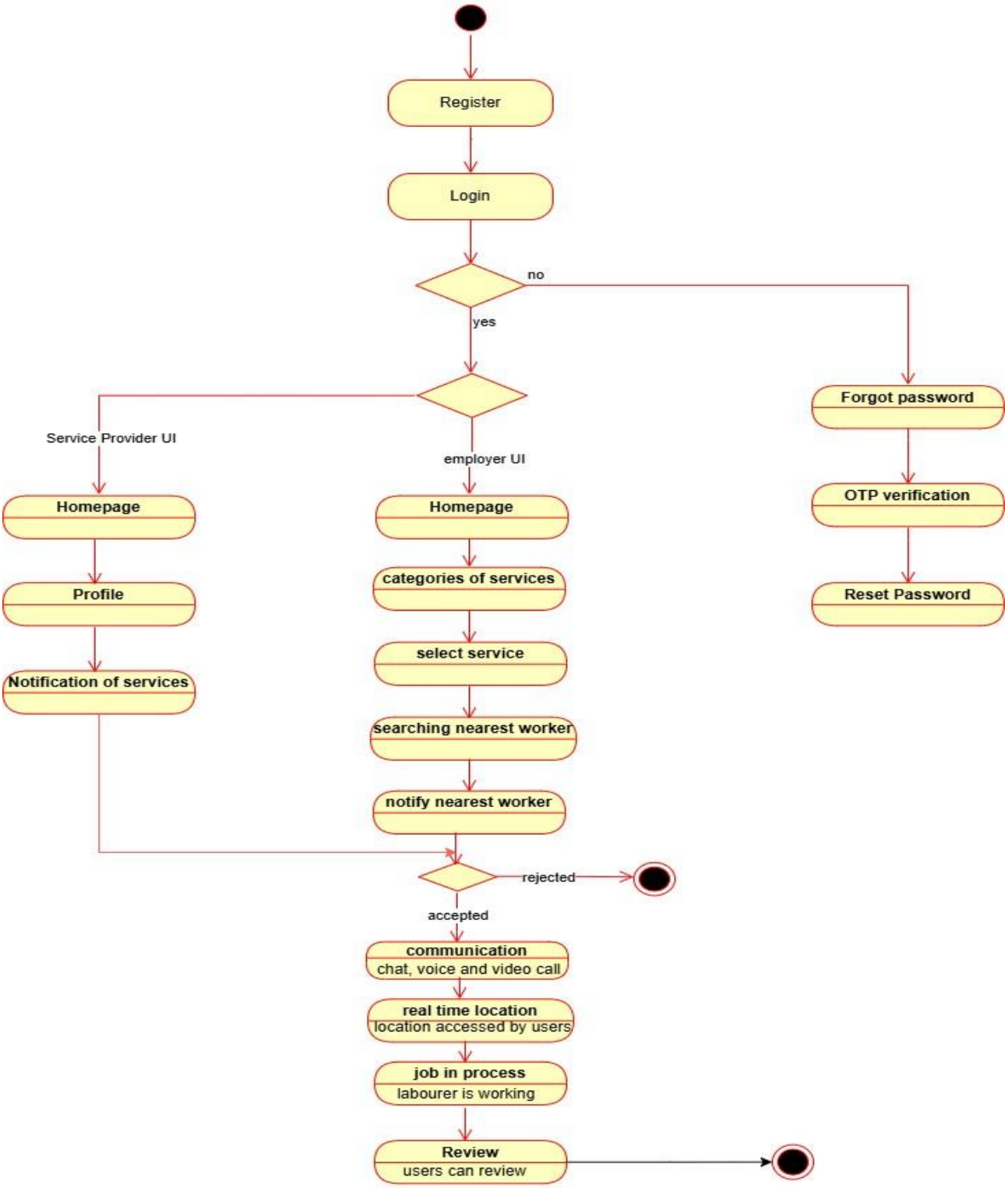


FIGURE 9 STATE TRANSITION DIAGRAM

Explanation of the State Transition Diagram:

The State Transition Diagram represents the flow of user actions within the **Chase Your Needs CYN** mobile app. Here is a summary of the states and transitions:

1. **Login/Sign Up:** The user must log in or sign up before accessing the app features.
2. **Homepage:** After successful login, the user is directed to the homepage where they can either.
3. **Explore Services:** Employers can seek services in the app to notify workers and select the category of work to be done.
4. **Notification:** Employer can notify, and service providers could respond to this notification.
5. **Communication:** Both users are connected to each other to explain the problem. Chat, voice, and video call options are available, so all kinds of users can use the application.
6. **Real-time location:** Both users' locations are shared with each other so that they can access the distance and exact arrival at work.
7. **Review:** Employers can review the service providers for work satisfaction.
8. **Reset Password:** If a user forgot his/her password, he could reset it easily by OTP verification.

This diagram showcases the simple flow of user interaction, highlighting the process from login to job completion

3.2.4 UML Component Diagram

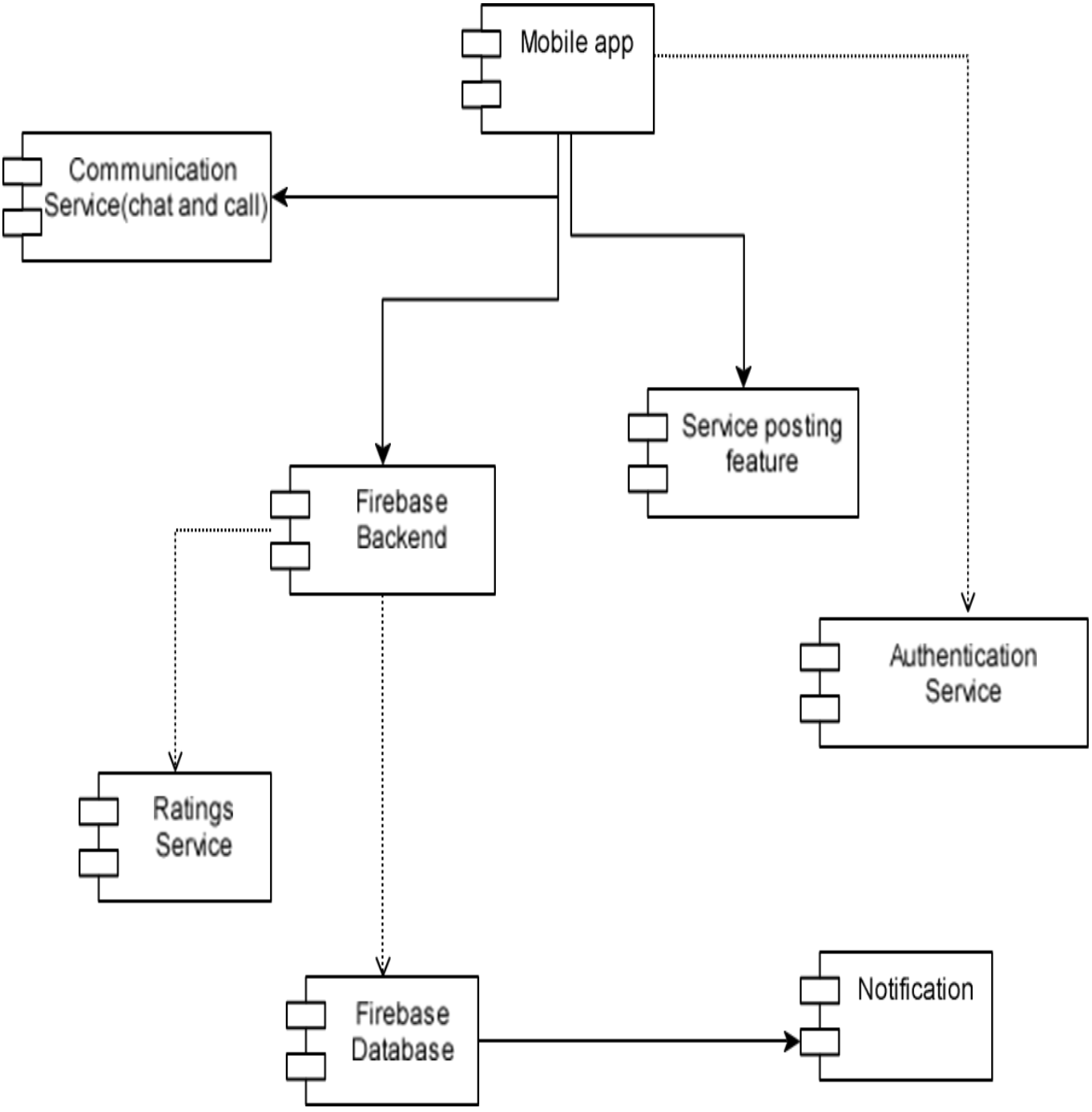


FIGURE 10 UML COMPONENT DIAGRAM

UML Component Diagram Explanation:

This UML Component Diagram represents the structure of a Service Provider and Employer Service access portal app.

1. **Mobile App & Authentication Service**

   Relationship: Dependency

   The Mobile App depends on the Authentication Service to authenticate users, i.e., the mobile app needs the authentication service for login and registration.

   Arrow: A dashed line with an open arrowhead pointing from Mobile App to Authentication Service.

2. **Mobile App & Firebase Backend:**

   Relationship: Association

   The Mobile App has a basic association with the Firebase Backend, meaning it communicates with the backend for various tasks such as posting Service, Communication (chat and call), and managing profiles.

   Arrow: A solid line connecting Mobile App and Firebase Backend.

3. **Firebase Backend & Firebase Database:**

   Relationship: Dependency

   The Firebase Backend depends on the Firebase Database to store and retrieve data (e.g., Services, user profiles, messages).

   Arrow: A dashed line with an open arrowhead pointing from Firebase Backend to Firebase Database.

4. **Firebase Backend & Notification Service:**

   Relationship: Association

   The Firebase Backend communicates with the Notification Service to send push notifications for service updates, chat messages, etc.

   Arrow: A solid line connecting Firebase Backend and Notification Service.

5. **Firebase Backend & Review & Rating Service:**

   Relationship: Dependency

   The Firebase Backend depends on the Review & Rating Service to manage reviews and ratings for completed works or services.

   Arrow: A dashed line with an open arrowhead pointing from Firebase Backend to Review & Rating Service

6. **Mobile App & Communication (Chat/Messaging/call)Service:**

Relationship: The Mobile App has an association with the communication Service since the app needs to allow users to send and receive messages and calls, but the app does not own the Communication service.

Arrow: A solid line connecting Mobile App and Communication Service.

7. **Mobile App & Service Posting Feature/Service:**

   Relationship: Association

   The Mobile has an association with Service Posting Services. A Service posting service allows Service Providers to post Service, and while the app can access this service, the Service posting service itself could potentially exist outside the app and is reusable.

   Arrow: A line with a solid line connecting Mobile App and Job Posting Service.

8. **Job Posting service & Firebase Database:**

   Relationship: Dependency

   The Job Posting Service depends on the Firebase Database to save and retrieve Service posts.

   Arrow: A dashed line with an open arrowhead pointing from Job Posting Service to Firebase Database.

Key Components & Roles:

Homepage – Central hub for navigation (login, Service listings, profile, chat).

Login/Signup – Handles user authentication and registration.

Service Posting – Allows Service providers to post Services and Employers to offer work to allow.

User Profile – Manages user details, passwords, and profile settings.

Chat – Facilitates messaging between Service Providers and Employers, including work offer notifications.

Component Interactions:

Homepage links to login, Service postings, and user profiles.

Service Posting and User Profile connect with Chat for communication.

Login/Signup ensures secure access.

3.2.5 Use Case Diagram

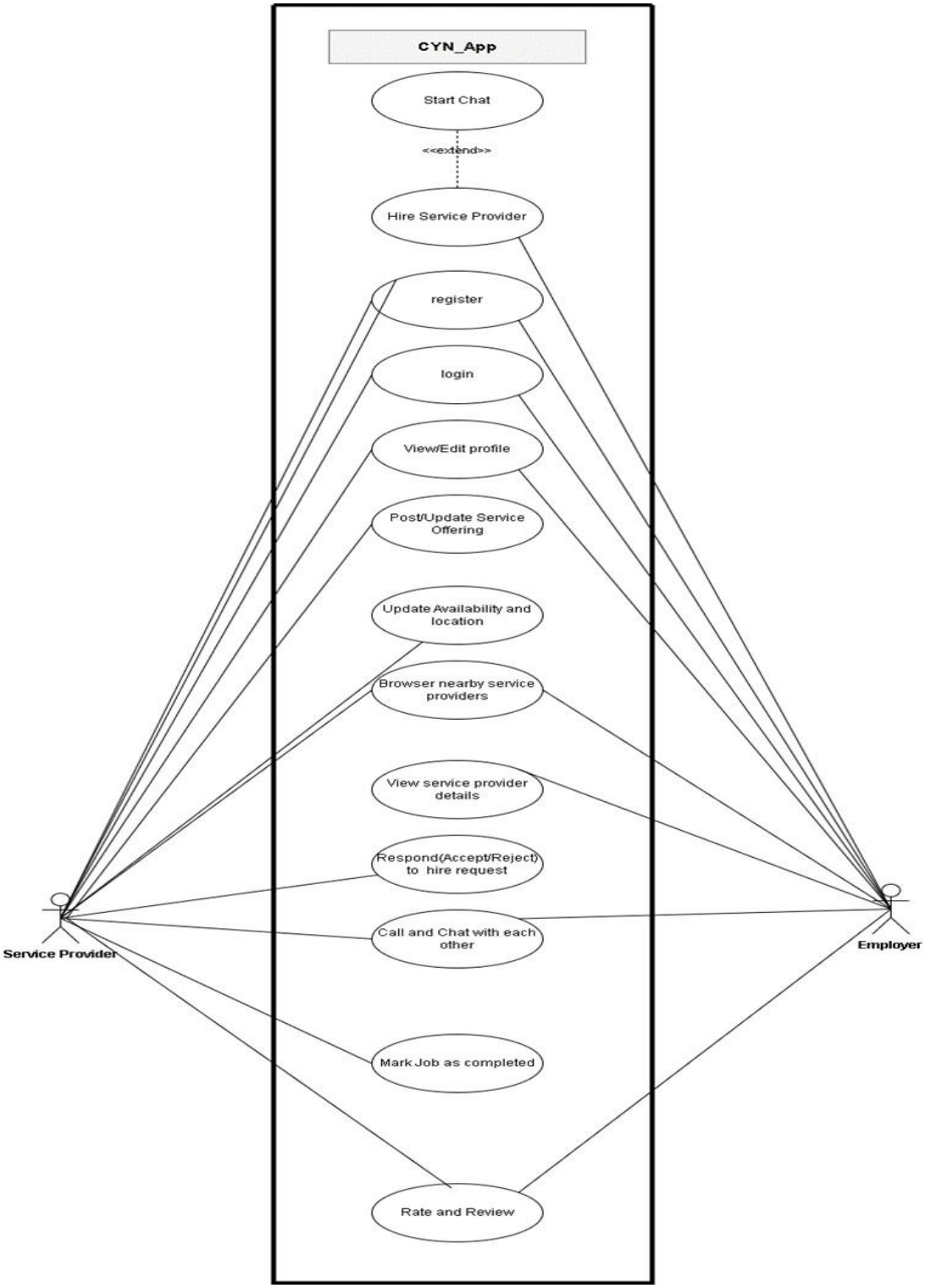


FIGURE 11 USE CASE DIAGRAM

Use Case Diagram Explanation:

The use case diagram illustrates how different actors interact with the service-based mobile application. It highlights the key functionalities available to customers, service providers, the admin, and the automated system.

1. **Customer /Employer (User):**
   - **Sign Up / Login:** The customer creates an account or logs into the app.
   - **Search for Services:** They browse or search for available services such as plumbing, electrical work, cleaning, etc.
   - **Book a Service:** After choosing a provider, the customer sends a service request.
   - **Track the Provider:** Once the provider accepts the request, the customer can view their live status or location.
   - **Rate & Review:** After the job is completed, the customer shares feedback about the provider.

2. **Service Provider**
   - **Register / Login:** Service providers create an account or log in.
   - **Accept or Reject Requests:** When a customer sends a request, the provider can choose to accept or decline it.
   - **Update Service Status:** Providers update their status as they move through the process for example, "On the Way," "Arrived", "In Progress," and "Completed."
   - **Edit Profile:** Providers can update their skills, availability, and pricing.

3. **System (Automated Processes)**
   - **Send Notifications:** Alerts customers and service providers about new requests, updates, payments, and reminders.
   - **Update Job Status:** Automatically updates stages of the service process based on provider actions.
   - **Track Providers:** Uses GPS to monitor and show the provider's live location to the customer.

Overall Summary

The diagram describes a smooth flow where customers search for services, providers respond to requests, the system automates updates and notifications. The interactions between all actors ensure the platform runs efficiently and reliably.

3.2.6 MVC Model

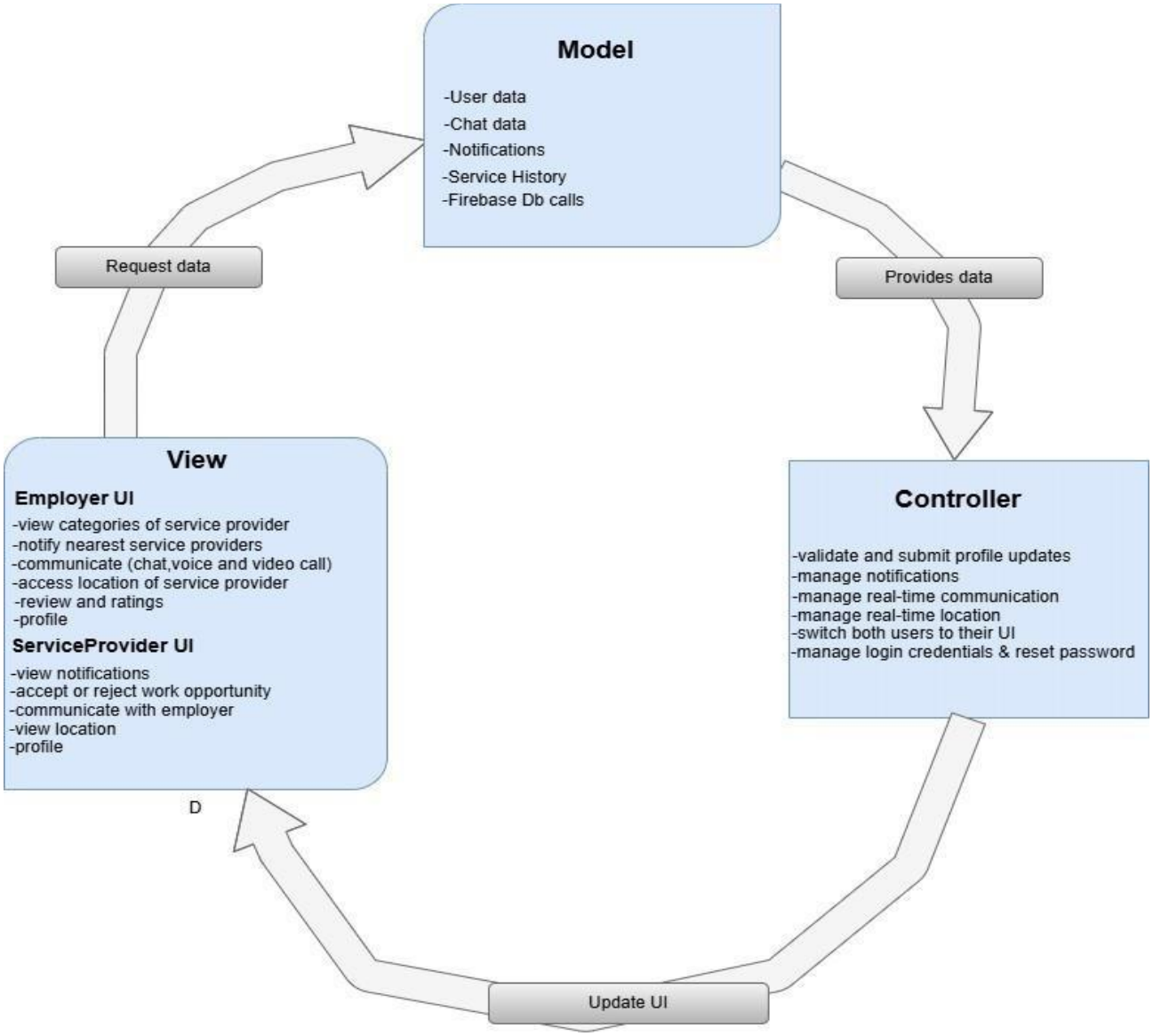


FIGURE 12 MVC MODEL

This diagram represents the Model-View-Controller (MVC) architecture for a Services access system:

Model: Handles data-related operations such as fetching user data, chat data, notifications, service history, and firebase database calls.

View: Manages the UI, displaying services categories, location, review and ratings, and user profile updates while handling errors.

Controller: Acts as an intermediary, processing user requests like work notifications, work approvals, location accessing and profile updates, then updating the UI accordingly.

Data flows from Model to View through the Controller, ensuring a structured and maintainable application.

3.2.7 ER Diagram

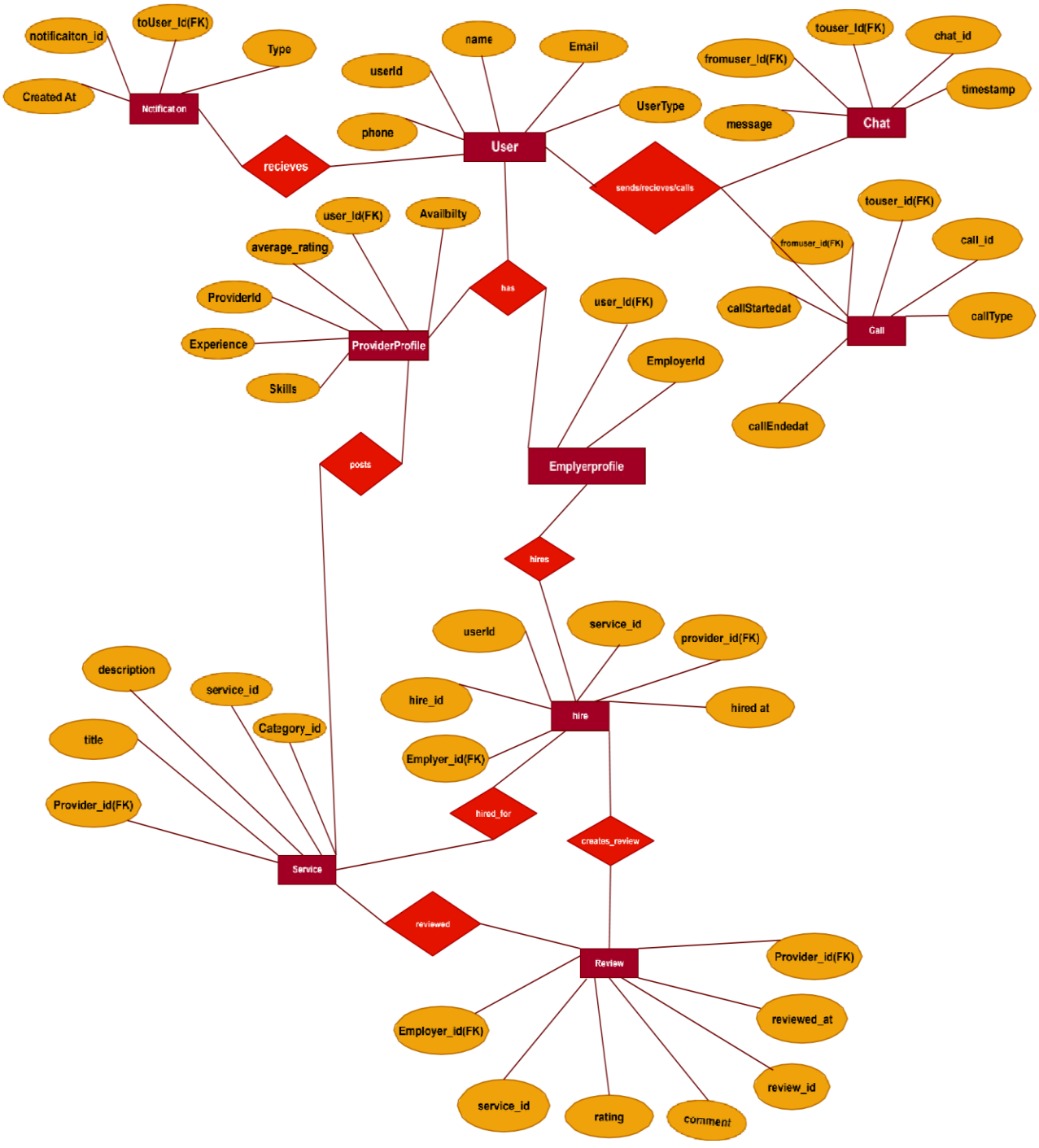


FIGURE 13 ER DIAGRAM

3.2.8 Behavioral Model

The behavioral model explains how the system behaves in response to user actions. It describes the flow of activities for customers and service providers.

CUSTOMER BEHAVIOR FLOW

1. The customer opens the application.
2. The customer logs in or creates a new account.
3. The customer selects the required service category.
4. The system accesses the customer's current location.
5. The customer submits a service request.
6. The system searches for nearby available providers.
7. The request is sent to the nearest provider.
8. If the provider accepts the request, the system creates a booking.
9. The chat and call options become available.
10. The customer can track the provider's arrival.
11. The provider completes the service.
12. The customer submits rating and review.

SERVICE PROVIDER BEHAVIOR FLOW

1. The service provider opens the application.
2. The provider logs in or creates a new account.
3. The provider sets availability status.
4. The provider receives a service request.
5. The provider views customer details, location, and problem description.
6. The provider accepts or rejects the request.
7. If accepted, the provider can chat or call the customer.
8. The provider travels to customer location.
9. The provider completes the service.
10. The provider receives rating and review from the customer.

REQUEST HANDLING BEHAVIOR

The request handling behavior is one of the most important parts of the CYN application.

1. Customer creates a service request.
2. System searches providers based on category and distance.
3. System sends request to the nearest available provider.
4. Provider receives request with a response timer.
5. Provider accepts or rejects the request.
6. If accepted, booking and chat are created.
7. If rejected, system forwards request to the next nearest provider.

8. If no provider is available, system displays a message to the customer.

CHAT BEHAVIOR

1. Chat is available only after request acceptance.
2. Customer and provider can send text messages.
3. Customer can send an image of the issue.
4. Provider can discuss required tools, material, time, and service charges.
5. Voice and video call options are available for better communication.

REVIEW BEHAVIOR

1. Review option appears after job completion.
2. Customer selects star rating.
3. Customer may write optional feedback.
4. System saves review in the database.
5. Provider's average rating is updated.
6. Review appears on provider profile.

3.3 Data Design

Data design explains how the application data will be organized and stored. Since CYN uses Firebase, the data can be stored in collections and documents. Each collection stores a specific type of data.

MAIN DATABASE COLLECTIONS

Collection Name	**Description**
users	Stores common user information.
Employer	Stores Employer-specific profile data.
Providers	Stores Service Provider profile and availability data.
categories	Stores service categories.
serviceRequests	Stores Employer/Customer service requests.
bookings	Stores accepted and completed jobs.
chats	Stores chat session information.
messages	Stores messages inside chat sessions.
reviews	Stores customer ratings and feedback.
notifications	Stores system notifications.

DATA DESIGN DESCRIPTION

When a new user creates an account, the user's basic information is stored in the users collection. If the user selects the customer role, additional customer data is stored in the customers collection. If the user selects the service provider role, provider details such as category, experience, rating, availability, and location are stored in the providers collection.

When a customer requests a service, a new document is created in the serviceRequests collection. If the provider accepts the request, the system creates a booking in the bookings collection. A chat session is also created for communication between the customer and provider. After the service is completed, the review is stored in the reviews collection.

3.3.1 DATA DICTIONARY

The data dictionary defines the fields used in the database and explains their data types and purpose.

Field Name	Data Type	Description
userId	String	Unique identifier of the user.
fullName	String	Full name of the user.
phone	String	Contact number of the user.
email	String	Email address of the user.
role	String	Role of user: Customer or Service Provider.
profileImage	String	URL of profile image.
createdAt	Timestamp	Account creation date and time.
customerId	String	Unique identifier of the customer.
providerId	String	Unique identifier of the service provider.
categoryId	String	Unique identifier of service category.
categoryName	String	Name of service category.
experience	String	Work experience of service provider.
rating	Double	Average rating of service provider.
totalJobs	Integer	Total completed jobs of provider.
availabilityStatus	Boolean	Shows whether provider is available.
latitude	Double	Latitude of user location.
longitude	Double	Longitude of user location.
address	String	User address or service location.
requestId	String	Unique identifier of service request.

problemDescription	String	Description of customer problem.
status	String	Status of request or booking.
bookingId	String	Unique identifier of booking.
serviceDate	Date	Date of service.
duration	String	Duration of completed service.
totalAmount	Double	Total service charges.
chatId	String	Unique identifier of chat session.
messageId	String	Unique identifier of message.
senderId	String	User ID of message sender.
messageText	String	Text content of message.
imageUrl	String	URL of image sent in chat.
messageType	String	Type of message: text, image, etc.
sentAt	Timestamp	Date and time when message was sent.
reviewId	String	Unique identifier of review.
reviewText	String	Customer feedback.
notificationId	String	Unique identifier of notification.
receiverId	String	User ID of notification receiver.
isRead	Boolean	Shows whether notification is read.

3.4 User Interface Design

The user interface of the CYN application is designed to be clean, modern, and easy to understand. The application uses a blue and white color theme to create a professional look. Cards, buttons, icons, and proper spacing are used to make the interface attractive and user-friendly.

The application interface is divided according to user roles. Customers have screens for searching providers, viewing profiles, chatting, tracking, and reviewing services. Service providers have screens for dashboard, request handling, job history, and profile management.

MAIN UI DESIGN GOALS

Goal	**Description**
Simplicity	The interface should be easy for all users.
Clarity	Buttons, icons, and text should be clear.
Fast Navigation	Users should move between screens easily.

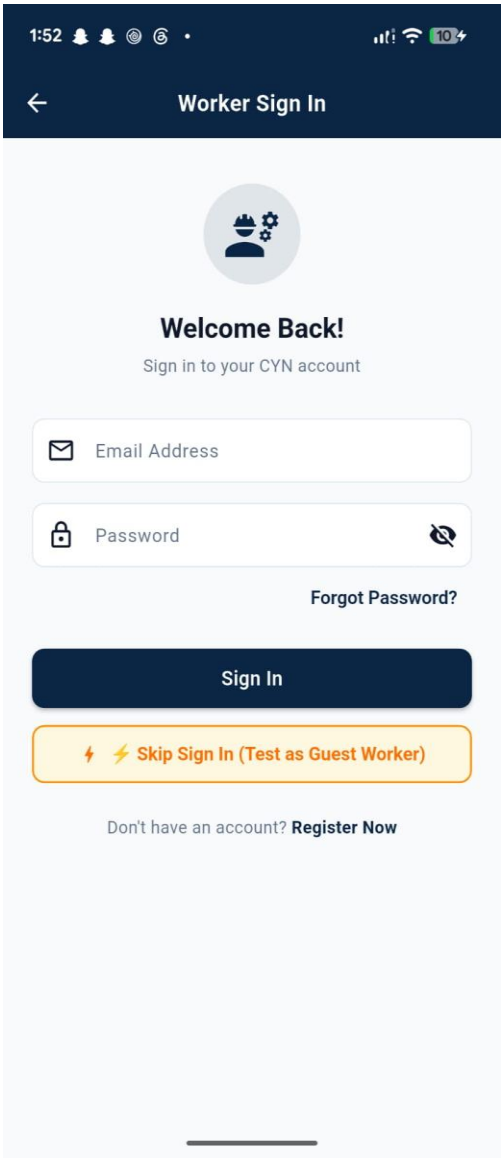
Consistency	Colors, fonts, buttons, and cards should follow the same style.
Accessibility	Important actions should be visible and easy to perform.
Responsiveness	Screens should work properly on different mobile screen sizes.

MAIN SCREENS OF THE APPLICATION

Screen Name	**Description**
Login Screen	Allows existing users to sign in using email/phone and password.
Create Account Screen	Allows users to create an account and select role.
Provider Dashboard	Shows provider availability, jobs done, rating, new requests, and recent jobs.
New Service Request Screen	Shows request details such as customer name, location, service type, and accept/reject buttons.
Customer Profile Screen	Shows customer personal details, booking history, saved addresses, and settings.
Searching for Providers Screen	Displays the searching process after customer sends a request.
Provider Profile Screen	Shows provider details, rating, experience, service category, reviews, chat, and call buttons.
Job Completed Screen	Allows customer to rate service provider and submit review.
Tracking Screen	Shows provider location, estimated arrival time, and contact options.
Chat Screen	Allows customer and provider to exchange messages and images.

3.4.1 SCREEN IMAGES

The screen images of the CYN application show the visual design and flow of the mobile application.


1:52
Worker Sign In
Welcome Back!
Sign in to your CYN account
Email Address
Password
Forgot Password?
Sign In
⚡ Skip Sign In (Test as Guest Worker)
Don't have an account? Register Now

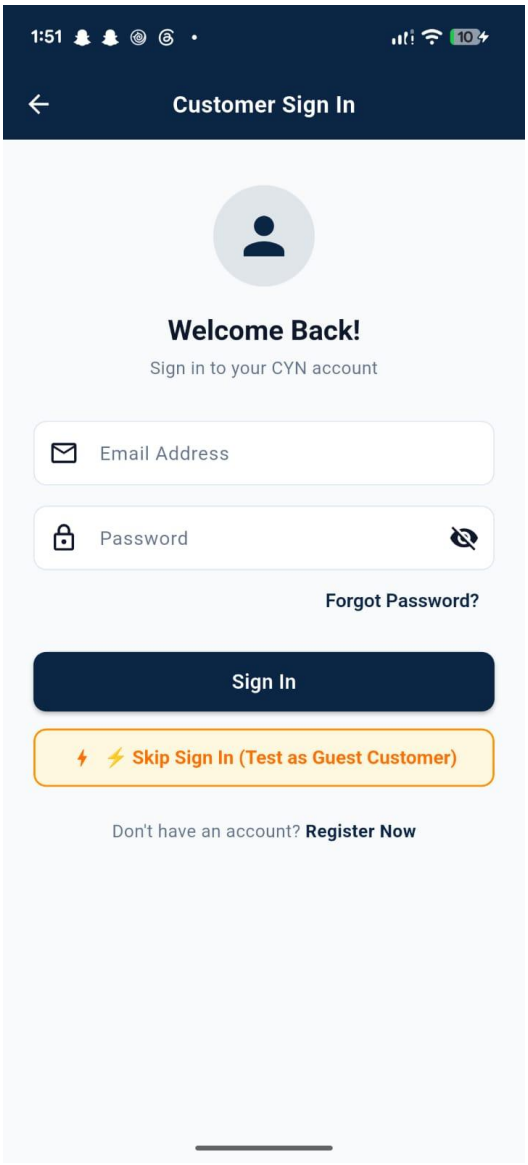
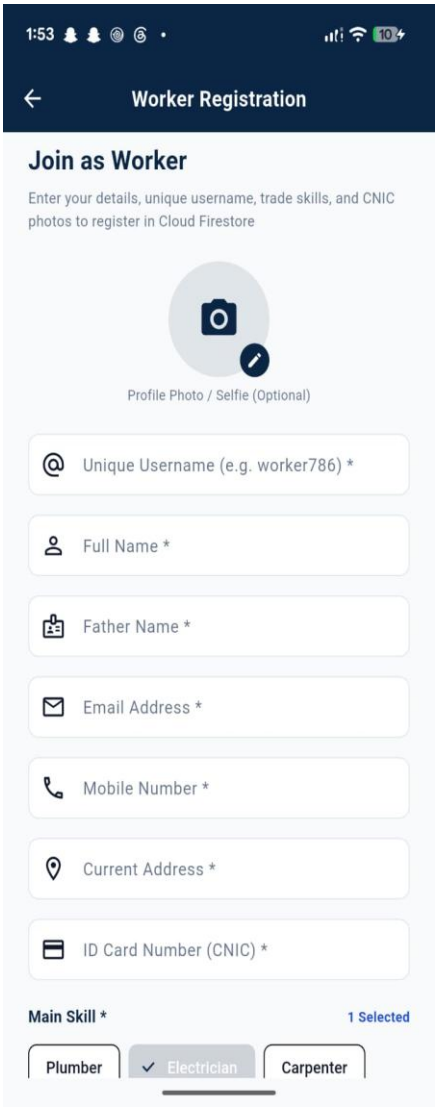


FIGURE 14 SIGN IN SCREENS


1:53
Worker Registration
Join as Worker
Enter your details, unique username, trade skills, and CNIC photos to register in Cloud Firestore
Profile Photo / Selfie (Optional)
Unique Username (e.g. worker786) *
Full Name *
Father Name *
Email Address *
Mobile Number *
Current Address *
ID Card Number (CNIC) *
Main Skill *
1 Selected
Plumber
Electrician
Carpenter

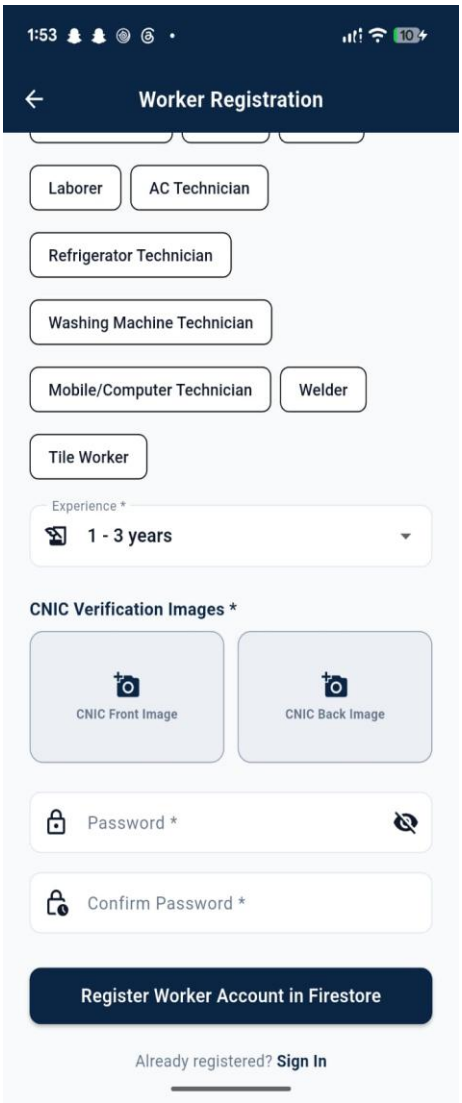
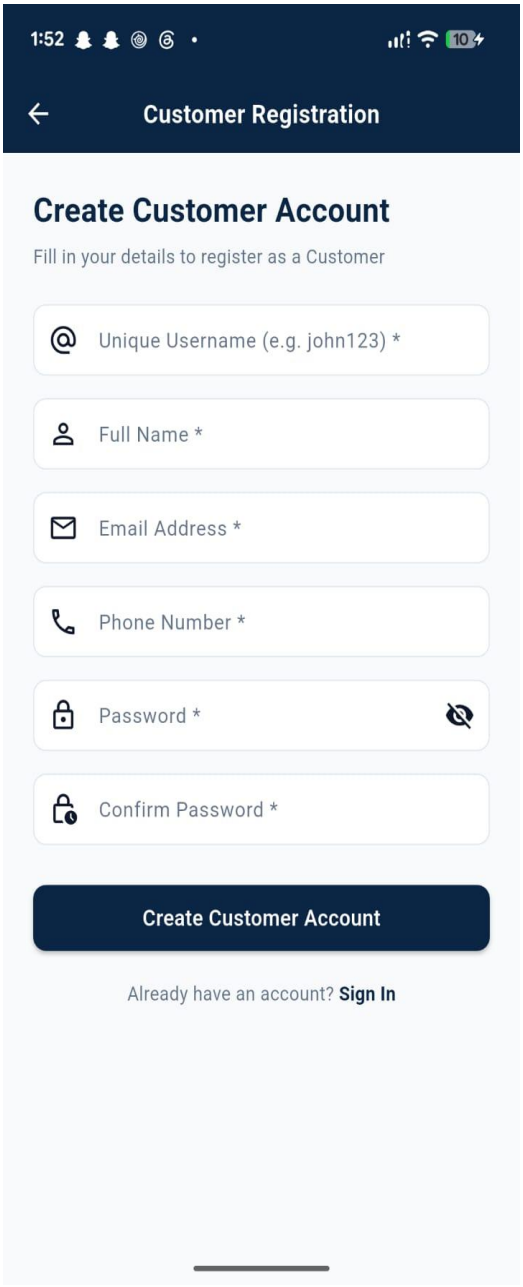


FIGURE 15 WORKER REGISTRATION SCREENS


1:52
Customer Registration
Create Customer Account
Fill in your details to register as a Customer
Unique Username (e.g. john123) *
Full Name *
Email Address *
Phone Number *
Password *
Confirm Password *
Create Customer Account
Already have an account? Sign In

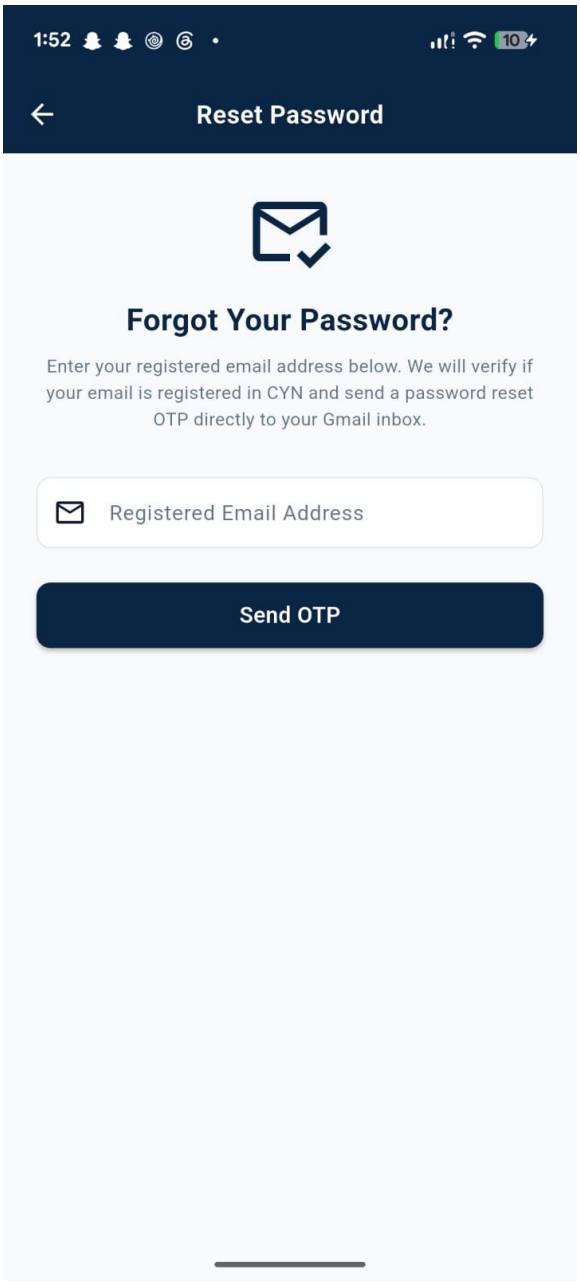


FIGURE 16 CUSTOMER REGISTRATION AND RESET PASSWORD SCREEN

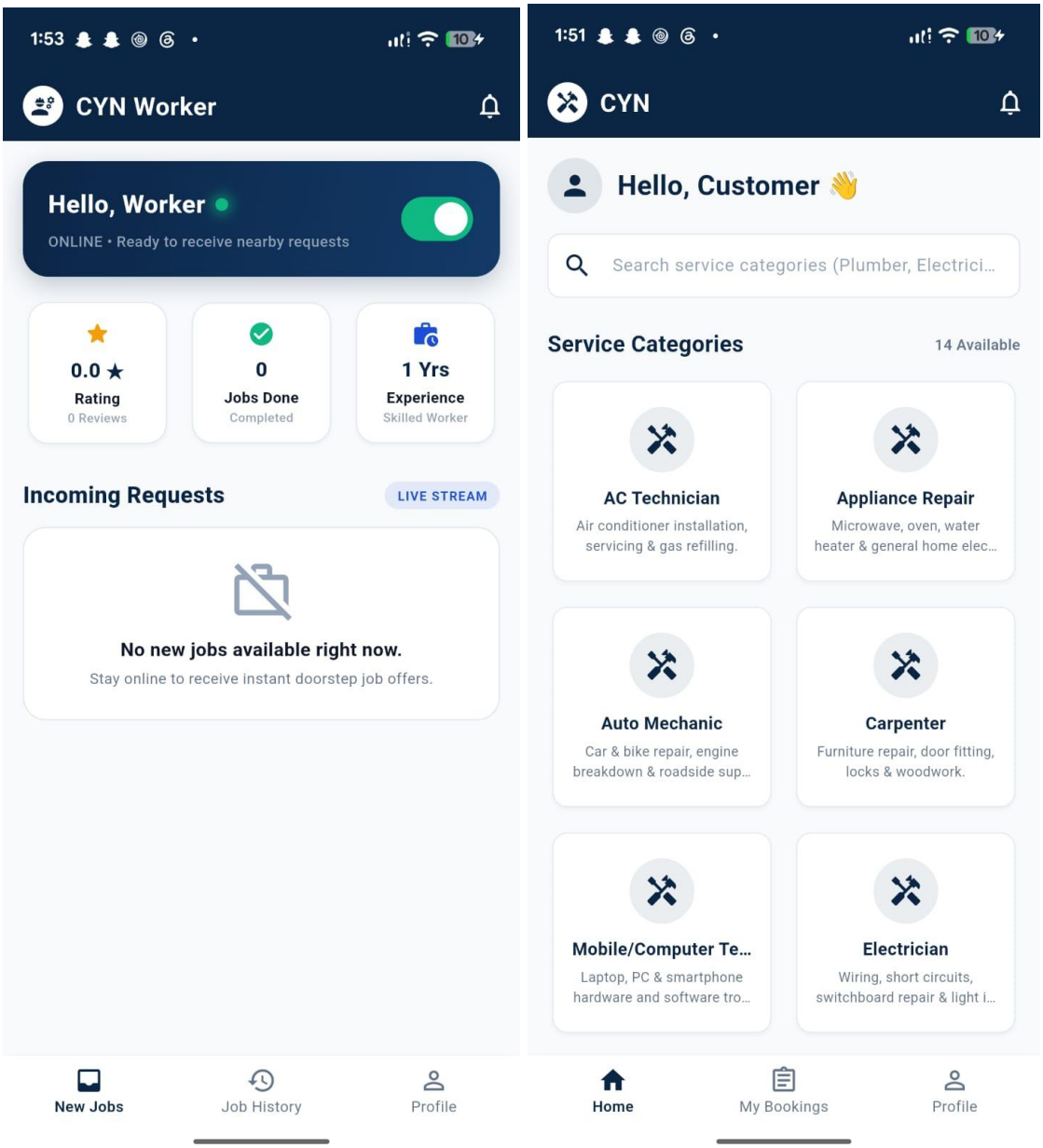


FIGURE 17 WORKER AND CUSTOMER DASHBOARDS

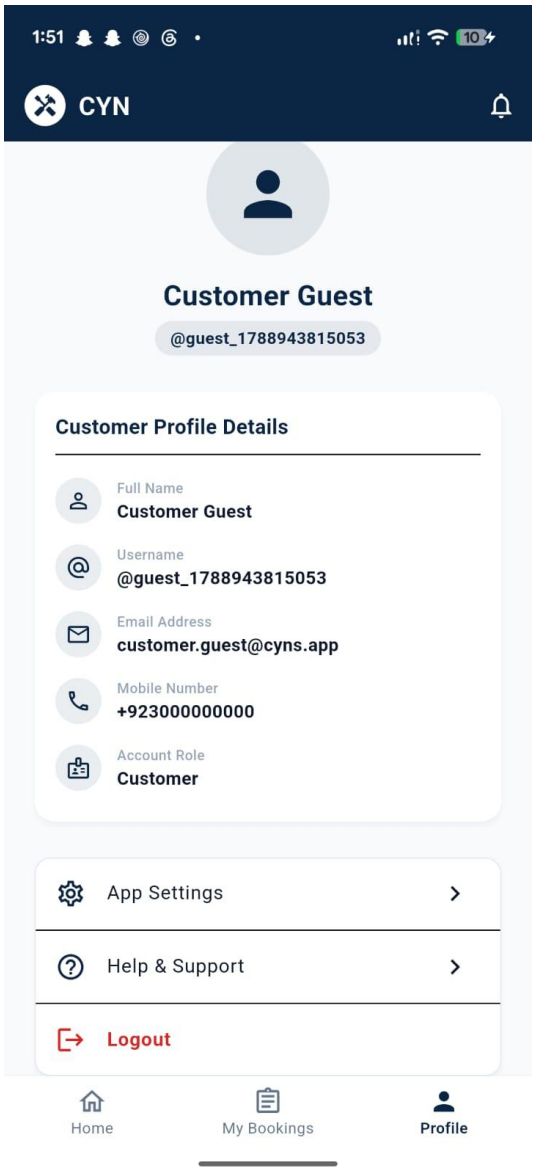


FIGURE 18 CUSTOMER PROFILE

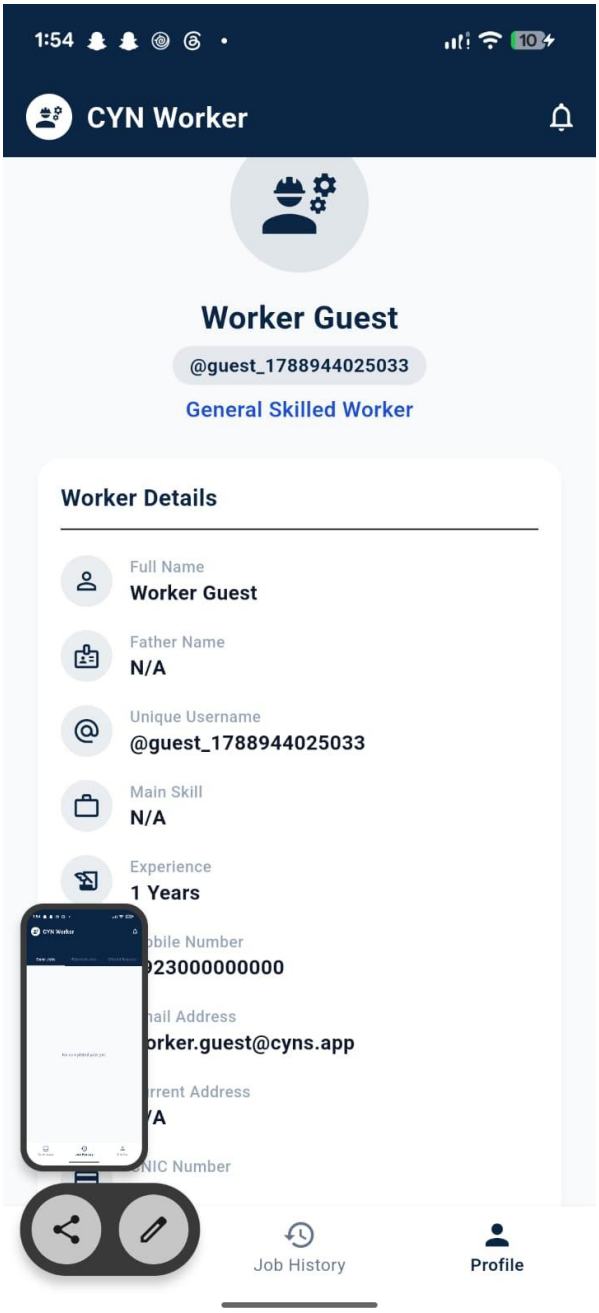


FIGURE 19 WORKER PROFILE

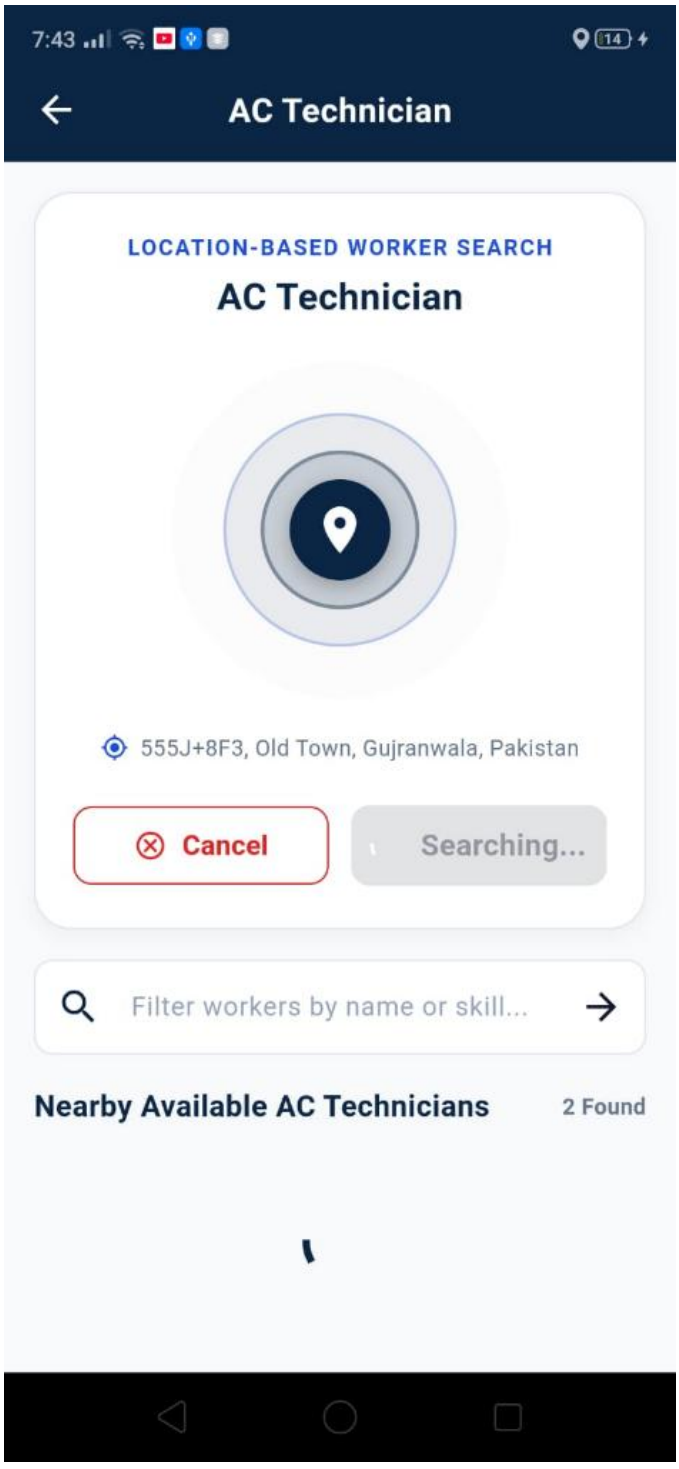


FIGURE 20 LOCATION BASED WORKER SEARCHING SCREEN

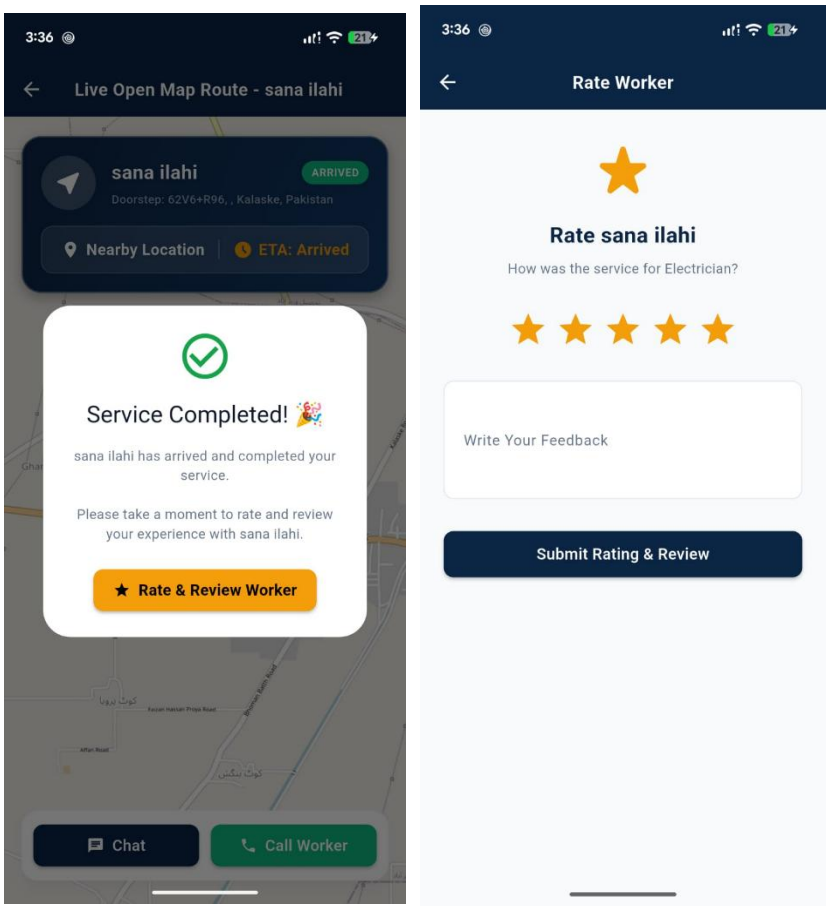


FIGURE 21 SERVICE COMPLETE AND RATE WORKER SCREEN

3.4.2 SCREEN OBJECTS AND ACTIONS

This section describes the main objects used on different screens and the actions performed by them.

Screen Object	Screen	Action
Email/Phone Text Field	Login Screen	Allows user to enter email or phone number.
Password Text Field	Login Screen	Allows user to enter password.
Login Button	Login Screen	Authenticates user and opens the relevant dashboard.
Create New Account Button	Login Screen	Opens the registration screen.
Full Name Text Field	Create Account Screen	Allows user to enter full name.
Phone Number Text Field	Create Account Screen	Allows user to enter phone number.
Role Selection Buttons	Create Account Screen	Allows user to select Customer or Service Provider.
Create Account Button	Create Account Screen	Creates a new user account.
Availability Toggle	Provider Dashboard	Allows provider to mark availability status.
View Request Button	Provider Dashboard	Opens new service request details.
Accept Button	Request Screen	Accepts the customer request and creates booking.
Reject Button	Request Screen	Rejects the request and ends provider session.
Chat Button	Provider Profile Screen	Opens chat module after request acceptance.
Call Button	Provider Profile Screen	Starts voice call between customer and provider.
Video Call Button	Chat Screen	Starts video call between customer and provider.
Send Message Button	Chat Screen	Sends text or image message.
Attachment Button	Chat Screen	Allows user to attach an image.
Cancel Request Button	Searching Screen	Cancels the active service request.
Message Button	Tracking Screen	Opens chat screen.

Submit Review Button	Job Completed Screen	Saves customer review and rating.
Logout Button	Profile Screen	Logs the user out of the application.

3.5 Design Decisions

The following design decisions were made for the CYN application:

Design Decision	**Reason**
Flutter was selected for mobile app development.	Flutter supports cross-platform development and provides a modern UI framework.
Firebase was selected as backend.	Firebase provides authentication, real-time database, storage, and notifications.
Location-based searching was included.	It allows customers to find the nearest available service provider.
Request forwarding was added.	It reduces waiting time when a provider rejects the request.
Provider availability status was included.	Requests should only be sent to providers who are free.
Chat module was included.	It allows customer and provider to discuss the problem before service.
Voice and video call options were included.	Calls help users explain complex problems more clearly.
Review and rating system was included.	It builds trust and improves provider profile reputation.
Blue and white theme was selected.	It gives the application a clean, professional, and modern look.
Firebase Cloud Messaging was selected.	It allows real-time notifications for requests and messages.
Firebase Storage was selected.	It supports storing profile images and issue images shared in chat.
Separate customer and provider flows were designed.	Both users have different needs and actions in the system.

3.6 Summary

This chapter presented the design and architecture of the Chase Your Need application. It explained the system design, design considerations, domain model, data flow diagram, ER diagram, architectural design, data design, data dictionary, user interface design, screen objects and actions, behavioral model, and design decisions.

The design of CYN is based on a mobile client-server architecture using Flutter and Firebase. The application allows customers to search for nearby service providers, send requests, communicate through chat or call, track providers, complete jobs, and submit reviews. Service providers can manage availability, accept or reject requests, communicate with customers, and build their profile through ratings and completed jobs.



Chapter 4: Implementation

4. Implementation

The implementation phase describes how the proposed design of the Chase Your Need (CYN) application is converted into a working mobile application. This phase focuses on the technologies, tools, algorithms, and development approach used to build the system.

CYN is planned to be developed as a mobile application using Flutter and Firebase. Flutter is used for designing the mobile user interface and developing the application logic. Firebase is used as the backend platform to manage authentication, database, notifications, cloud storage, and real-time communication.

The application contains two main user roles: Customer and Service Provider. Customers can create accounts, log in, select service categories, search for nearby available workers, send service requests, communicate through chat or calls, track service providers, and submit reviews after service completion. Service providers can register themselves, manage their profiles, update availability status, receive service requests, accept or reject offers, communicate with customers, complete jobs, and view ratings.

The main implementation goal of the CYN application is to provide a fast, reliable, and user-friendly platform where customers can easily hire nearby service providers.

4.1 Algorithm

The most important algorithm in the CYN application is the Nearest Available Service Provider Search Algorithm. This algorithm is used when a customer selects a service category and sends a request. The system searches for nearby available providers and sends the request to the nearest worker.

If the nearest service provider accepts the request, the system creates a booking and opens the chat module. If the provider rejects the request or does not respond, the system forwards the request to the next nearest available provider.

NEAREST AVAILABLE PROVIDER SEARCH ALGORITHM

- **Step 1:** Customer logs in to the application.
- **Step 2:** Customer selects a required service category, such as electrician, carpenter, mechanic, mason, labor, loader rickshaw, painter, or plumber.
- **Step 3:** The system gets the customer's current location using GPS.
- **Step 4:** The system fetches all service providers related to the selected category from the database.
- **Step 5:** The system filters only those providers who are currently available.
- **Step 6:** The system calculates the distance between the customer and each available provider.
- **Step 7:** The providers are sorted according to the nearest distance.
- **Step 8:** The service request is sent to the nearest available provider.
- **Step 9:** The provider can accept or reject the request.
- **Step 10:** If the provider accepts the request, the system creates a booking and opens chat, call, and tracking options.

- **Step 11:** If the provider rejects the request, the system forwards the request to the next nearest available provider.
- **Step 12:** If no provider is available, the system displays a message to the customer.

DISTANCE CALCULATION

The system uses the latitude and longitude of the customer and service provider to calculate the distance between them. The provider with the shortest distance is considered the nearest available provider.

The distance can be calculated using a location-based distance formula or by using a location service package. The calculated distance helps the system arrange providers from nearest to farthest.

REQUEST ACCEPTANCE ALGORITHM

This algorithm handles the response of the service provider after receiving a request.

- **Step 1:** Provider receives a new service request notification.
- **Step 2:** Provider views customer name, location, service category, and problem description.
- **Step 3:** Provider chooses either Accept or Reject.
- **Step 4:** If provider accepts, request status changes to accepted.
- **Step 5:** A booking record is created.
- **Step 6:** Chat and call options are enabled.
- **Step 7:** If provider rejects, request status changes to rejected.
- **Step 8:** The request is forwarded to the next nearest available provider.

CHAT AND CALL FLOW ALGORITHM

- **Step 1:** The provider must accept request.
- **Step 2:** System creates a chat session between customer and provider.
- **Step 3:** Customer and provider can send text messages.
- **Step 4:** Customer can send an image of the problem if needed.
- **Step 5:** Voice and video call options become available.
- **Step 6:** Customer and provider can discuss required material, tools, time, and service fee.
- **Step 7:** Chat history is stored in the database.

REVIEW SUBMISSION ALGORITHM

- **Step 1:** Service provider completes the job.
- **Step 2:** Booking status changes to completed.
- **Step 3:** Customer receives review option.
- **Step 4:** Customer selects star rating.
- **Step 5:** Customer writes feedback if required.
- **Step 6:** System saves review in the database.
- **Step 7:** Provider's average rating is updated.

-
- **Step 8:** Review appears on the provider profile.

4.2 External APIs/SDKs

The CYN application uses different APIs, SDKs, frameworks, and packages to implement its features. These external tools help in mobile app development, backend management, authentication, real-time database operations, notifications, storage, location tracking, and communication.

API / SDK / Tool	Purpose
Flutter SDK	Used for developing the mobile application for Android and iOS using a single codebase.
Dart	Programming language used to write Flutter application code.
Firebase Authentication	Used for secure user registration and login.
Cloud Firestore	Used to store users, providers, service requests, bookings, chats, reviews, and notifications.
Firebase Realtime Database	Can be used for real-time status updates, chat, and tracking if required.
Firebase Cloud Messaging	Used to send push notifications for requests, messages, booking updates, and job completion.
Firebase Storage	Used to store profile images, service issue images, and other media files.
Google Maps SDK	Used to display maps and provider location.
Geolocator Package	Used to get the current location of customer and service provider.
Geocoding Package	Used to convert latitude and longitude into readable address format.
Image Picker Package	Used to select or capture images from the camera or gallery.
Permission Handler Package	Used to request mobile permissions such as location, camera, and microphone.
Voice/Video Call SDK	Used to provide voice and video call communication between customer and provider.
Shared Preferences	Used to store small local data such as login session or user role.
Flutter Local Notifications	Used to show local notifications inside the mobile application.

FIREBASE SERVICES IMPLEMENTATION

Firebase plays an important role in the implementation of the CYN application. The following Firebase services are used:

1. Firebase Authentication

Firebase Authentication is used to manage user registration and login. It allows customers and service providers to create accounts and securely access the application.

Main functions include:

- User signup
- User login
- Password authentication
- User session management
- Role-based access

2. Cloud Firestore

Cloud Firestore is used as the main database of the application. It stores structured data related to users, providers, categories, service requests, bookings, chats, reviews, and notifications.

Main collections include:

- users
- customers
- providers
- categories
- serviceRequests
- bookings
- chats
- messages
- reviews
- notifications

3. Firebase Cloud Messaging

Firebase Cloud Messaging is used to send push notifications to users. Notifications are sent when:

- A new service request is created
- A provider accepts a request
- A provider rejects a request
- A new chat message is received
- A job is completed
- A review is submitted

4. Firebase Storage

Firebase Storage is used to store media files such as profile images and images shared in chat. For example, a customer can send an image of a damaged item or service issue to the provider.

FLUTTER IMPLEMENTATION

Flutter is used to develop the mobile application interface and handle user interaction. It provides reusable widgets, smooth navigation, and a modern design structure.

Main Flutter implementation areas include:

- Login and signup screens
- Customer home screen
- Service category selection
- Provider dashboard
- Request screen
- Provider profile screen
- Chat screen
- Tracking screen
- Review screen
- Profile management screen

Flutter also connects with Firebase services through Firebase packages. The application sends and receives data from Firebase and updates the user interface in real time.

LOCATION IMPLEMENTATION

Location services are used to find nearby service providers. The application gets the current location of the customer and service providers using GPS. The system then compares the location of the customer with available providers and selects the nearest provider.

Location implementation includes:

- Requesting location permission
- Getting current latitude and longitude
- Storing provider location in database
- Calculating distance between customer and provider
- Showing provider arrival status
- Displaying estimated arrival information

NOTIFICATION IMPLEMENTATION

Notifications are used to keep users updated about important events in the application. When a customer sends a service request, the provider receives a notification. Similarly, the customer receives notifications when the provider accepts or rejects the request.

Notification events include:

- New service request
- Request accepted
- Request rejected

- New message received
- Provider arriving
- Job completed
- Review submitted

SECURITY IMPLEMENTATION

Security is implemented using Firebase Authentication and database security rules. Only authenticated users can access the application features. Customers and providers can only access data related to their own account and service records.

Security features include:

- Secure login and signup
- User role verification
- Protected database access
- Secure storage of user information
- Permission-based access to location, camera, and microphone

SUMMARY OF IMPLEMENTATION

The implementation of the CYN application is based on Flutter and Firebase. Flutter is used to build the mobile application interface, while Firebase is used to manage backend services. The main implemented features include authentication, service category selection, location-based provider search, request handling, chat, voice/video call, tracking, job completion, notifications, and review system.

The algorithms used in the system ensure that customers are connected with the nearest available service providers. The use of external APIs and SDKs helps make the application real-time, secure, and user-friendly.



Chapter 5: Testing and Evaluation

5. Introduction

Testing and evaluation are important phases in the development of the Chase Your Need (CYN) application. The purpose of testing is to verify that the system works according to the defined requirements and provides correct results to users. Testing also helps identify errors, improve system quality, and ensure that the application is reliable, secure, and easy to use.

CYN is a mobile application that connects customers with nearby service providers such as electricians, carpenters, auto mechanics, masons, labor workers, loader rickshaw drivers, painters, plumbers, and other related workers. Since the application includes features such as user registration, login, location-based provider search, request handling, chat, calls, tracking, and review system, proper testing is required to make sure all modules work correctly.

The application will be tested using different testing techniques, including:

- Unit Testing
- Functional Testing
- Integration Testing
- Performance Testing

Each testing method checks the system from a distinct perspective. Unit testing checks individual modules, functional testing checks system features, integration testing checks communication between modules, and performance testing checks system speed and stability.

5.1 Unit Testing (UT)

Unit testing is used to test individual units or modules of the application separately. Each function or module is tested to make sure it performs its task correctly. In the CYN application, unit testing is performed on modules such as login, signup, category selection, location access, request creation, chat message sending, and review submission.

The main purpose of unit testing is to detect errors at an early stage before combining different modules.

UNIT TESTING OBJECTIVES

The objectives of unit testing are:

- To test individual functions separately.
- To verify that each module gives the expected output.
- To identify errors in small parts of the system.
- To make debugging easier.
- To ensure that modules are ready for integration.

UNIT TEST CASES

Test Case ID	Module	Test Description	Input	Expected Result	Status
UT-01	Login	Test login with valid credentials	Correct email/phone and password	User logs in successfully	Pass
UT-02	Login	Test login with invalid password	Correct email/phone and wrong password	Error message is displayed	Pass
UT-03	Signup	Test account creation with valid data	Name, phone, email, password, role	Account is created successfully	Pass
UT-04	Signup	Test signup with missing required fields	Empty name or phone field	Validation message is displayed	Pass
UT-05	Role Selection	Test customer role selection	Customer role selected	Customer account type is saved	Pass
UT-06	Role Selection	Test service provider role selection	Provider role selected	Provider account type is saved	Pass
UT-07	Category Selection	Test service category selection	Electrician category selected	Selected category is saved	Pass
UT-08	Location Access	Test location permission	Location permission allowed	Current location is fetched	Pass
UT-09	Request Creation	Test service request creation	Category, location, problem description	Request is stored in database	Pass
UT-10	Chat	Test message sending	Text message	Message is sent and stored	Pass
UT-11	Review	Test review submission	Rating and feedback	Review is saved successfully	Pass
UT-12	Provider Availability	Test availability toggle	Available/Unavailable selected	Provider status is updated	Pass

UNIT TESTING RESULT

The unit testing results show that individual modules of the CYN application perform their functions correctly. Basic validations, data saving, user actions, and module-level operations work as expected.

5.2 Functional Testing (FT)

Functional testing is used to verify that the application features work according to the functional requirements. It checks the complete behavior of the system from the user's point of view.

In the CYN application, functional testing ensures that customers can search for service providers, send requests, communicate with providers, track provider arrival, and submit reviews. It also ensures that service providers can receive requests, accept or reject them, manage availability, and complete jobs.

FUNCTIONAL TESTING OBJECTIVES

The objectives of functional testing are:

- To verify that all major features work correctly.
- To check that user actions produce expected results.
- To ensure that customer and provider flows are working properly.
- To validate the system against functional requirements.

FUNCTIONAL TEST CASES

Test Case ID	Feature	Test Scenario	Expected Result	Status
FT-01	User Registration	Customer creates a new account	Customer account is created and customer dashboard opens	Pass
FT-02	User Registration	Service provider creates a new account	Provider account is created and provider dashboard opens	Pass
FT-03	User Login	Registered user logs in	User is redirected to relevant dashboard	Pass
FT-04	Service Category	Customer selects a service category	Selected category is processed	Pass
FT-05	Provider Search	Customer searches for nearby electrician	System searches available electricians near customer location	Pass
FT-06	Send Request	Customer sends service request	Request is sent to nearest available provider	Pass
FT-07	Accept Request	Provider accepts request	Booking is created and chat is enabled	Pass
FT-08	Reject Request	Provider rejects request	Request is forwarded to next nearest provider	Pass
FT-09	Chat Module	Customer sends message to provider	Provider receives message successfully	Pass
FT-10	Voice Call	Customer selects voice call option	Voice call screen opens	Pass
FT-11	Video Call	Customer selects video call option	Video call screen opens	Pass
FT-12	Tracking	Customer views provider arrival status	Provider location/status is displayed	Pass
FT-13	Job Completion	Provider marks job as completed	Booking status changes to completed	Pass
FT-14	Review System	Customer submits rating and review	Review is saved and provider rating updates	Pass
FT-15	Notifications	Provider receives new request notification	Notification is displayed to provider	Pass
FT-16	Profile Management	User updates profile information	Updated profile information is saved	Pass

FUNCTIONAL TESTING RESULT

Functional testing confirms that the primary features of the CYN application work according to the defined requirements. The customer and service provider flows are successfully tested, and the system performs the required functions properly.

5.3 Integration Testing (IT)

Integration testing is performed after unit testing. It checks whether different modules of the system work correctly when combined. In the CYN application, many modules depend on each other. For example, the request module depends on user authentication, location access, provider availability, notification, and database modules.

Integration testing ensures that data flows correctly between different parts of the system.

INTEGRATION TESTING OBJECTIVES

The objectives of integration testing are:

- To check communication between different modules.
- To verify that combined modules work correctly.
- To detect interface errors between modules.
- To ensure smooth data transfer between Flutter app and Firebase backend.
- To verify complete user flow from request creation to review submission.

INTEGRATION TEST CASES

Test Case ID	Integrated Modules	Test Description	Expected Result	Status
IT-01	Authentication + Dashboard	User logs in and system checks role	Correct dashboard opens according to user role	Pass
IT-02	Signup + Firebase Database	New user registers and data is saved	User record is stored in database	Pass
IT-03	Category + Location Search	Customer selects category and location is fetched	Nearby providers are searched	Pass
IT-04	Location + Provider Database	System compares customer and provider location	Nearest available provider is selected	Pass
IT-05	Request + Notification	Customer sends request to provider	Provider receives request notification	Pass
IT-06	Request + Provider Response	Provider accepts request	Request status changes to accepted	Pass
IT-07	Request + Forwarding Logic	Provider rejects request	Request is sent to next nearest provider	Pass
IT-08	Booking + Chat	Accepted request creates booking and chat	Chat session is created successfully	Pass
IT-09	Chat + Firebase	User sends message	Message is stored and displayed in real time	Pass
IT-10	Booking + Review	Customer submits review after completed job	Review is linked with booking and provider	Pass
IT-11	Provider Availability + Request	Provider is unavailable	Request is not sent to unavailable provider	Pass
IT-12	Notification + Chat	New message is sent	Receiver gets message notification	Pass

INTEGRATION TESTING RESULT

Integration testing shows that different modules of the CYN application work correctly together. The system successfully handles login, provider search, request forwarding, chat creation, notification delivery, booking completion, and review submission.

5.4 Performance Testing (PT)

Performance testing is used to check the speed, response time, stability, and efficiency of the application. Since CYN uses real-time features such as provider search, chat, notification, and tracking, performance testing is necessary to ensure that the system responds quickly.

The application should perform smoothly on different mobile devices and should handle multiple users without crashing.

PERFORMANCE TESTING OBJECTIVES

The objectives of performance testing are:

- To check application response time.
- To verify smooth screen loading.
- To test real-time chat performance.
- To check notification delivery speed.
- To test provider search speed.
- To ensure system stability under multiple requests.

PERFORMANCE TEST CASES

Test Case ID	Test Scenario	Expected Result	Status
PT-01	Application launch time	Application should open within acceptable time	Pass
PT-02	Login response time	User should log in without long delay	Pass
PT-03	Provider search speed	Nearby provider search should complete quickly	Pass
PT-04	Request delivery time	Service request should reach provider quickly	Pass
PT-05	Chat message delivery	Message should appear in real time	Pass
PT-06	Notification delivery	Notification should be delivered without major delay	Pass
PT-07	Map/location loading	Location and tracking screen should load smoothly	Pass
PT-08	Multiple requests	System should handle multiple service requests	Pass
PT-09	Profile image loading	Profile image should load properly	Pass
PT-10	Review submission	Review should save without delay	Pass

PERFORMANCE TESTING RESULT

The performance testing results show that the CYN application provides acceptable response time and smooth user experience. The system can perform major operations such as login, provider search, request delivery, chat messaging, and review submission efficiently.

5.5 Summary

This chapter presented the testing and evaluation of the Chase Your Need application. Different testing techniques were discussed, including unit testing, functional testing, integration testing, and performance testing.

Unit testing verified individual modules such as login, signup, category selection, request creation, chat, and review submission. Functional testing ensured that the main features of the system work according to requirements. Integration testing checked the communication between different modules such as authentication, location, request, booking, chat, notification, and review. Performance testing evaluated the speed, response time, and stability of the application.

Overall, the testing process confirms that the CYN application is reliable, functional, efficient, and ready for further deployment and user evaluation.



Chapter 6: System Conversion

6. Introduction

System conversion is the process of changing from the existing manual system to the newly developed digital system. In the existing system, customers search for local service providers through personal contacts, references, phone calls, or by visiting nearby markets. This process is time-consuming and does not always provide trusted or available workers.

The Chase Your Need (CYN) application converts this manual process into a mobile-based digital system. Through this application, customers can search for nearby service providers such as electricians, carpenters, auto mechanics, masons, labour workers, loader rickshaw drivers, painters, plumbers, and other related workers.

The system allows customers to select a service category, search nearby available providers, send service requests, communicate through chat or call, track provider arrival, and submit reviews after service completion. Service providers can receive job requests, accept or reject them, manage availability, communicate with customers, and build their profiles through ratings and completed jobs.
The purpose of system conversion is to ensure that the new CYN application is introduced smoothly and users can shift from the manual hiring method to the digital platform without difficulty.

6.1 Conversion Method

For the CYN application, the recommended conversion method is Pilot Conversion.
Pilot conversion means that the system is first introduced to a limited number of users or in a small selected area before launching it on a larger scale. This method is suitable for CYN because the application involves real users, service providers, location tracking, service requests, communication, and reviews. Testing the system with a small group first helps identify problems before full deployment.

REASONS FOR SELECTING PILOT CONVERSION

Pilot conversion is selected for the following reasons:

Reason	Description
Low Risk	The application can be tested with limited users before full launch.
Easy Feedback Collection	Customers and service providers can provide feedback during initial use.
Error Detection	Bugs and issues can be identified before large-scale deployment.
User Training	Service providers and customers can be trained gradually.
Better Improvement	The system can be improved based on real user experience.
Controlled Deployment	The project team can monitor system performance in a limited environment.

PILOT CONVERSION PROCESS

The pilot conversion process for CYN will include the following steps:

1. Select a small area or limited group of users for initial testing.
2. Register selected customers and service providers.
3. Add basic service categories such as electrician, plumber, carpenter, painter, mechanic, mason, labour, and loader rickshaw.
4. Test service request flow from customer to provider.
5. Test request acceptance, rejection, and forwarding.
6. Test chat, voice call, video call, tracking, and review features.
7. Collect feedback from customers and service providers.
8. Fix issues and improve system performance.
9. Launch the application for a wider audience after successful pilot testing.

6.2 Deployment

Deployment is the process of making the application available for actual users. After development and testing, the CYN mobile application will be deployed so that customers and service providers can use it in a real environment.

The application will be developed using Flutter, while Firebase will be used as the backend service. Before deployment, Firebase Authentication, Firestore Database, Firebase Storage, Firebase Cloud Messaging, and location services must be properly configured.

DEPLOYMENT STEPS

The deployment process of the CYN application will include the following steps:

Step No.	Deployment Step	Description
1	Complete Development	Finalize all application modules including login, signup, provider search, request handling, chat, tracking, and review.
2	Configure Firebase	Set up Firebase Authentication, Firestore Database, Firebase Storage, and Firebase Cloud Messaging.
3	Set Database Rules	Apply security rules to protect user data and control access.
4	Test Application	Perform unit, functional, integration, and performance testing.
5	Build Application	Generate Android APK or App Bundle from Flutter project.
6	Pilot Deployment	Launch the application for selected users and service providers.
7	Collect Feedback	Take feedback from users about usability, performance, and features.
8	Fix Issues	Resolve bugs, improve UI, and optimize performance.
9	Final Deployment	Release the final version for wider use.

DEPLOYMENT ENVIRONMENT

The deployment environment for CYN will include:

Component	Description
Mobile Platform	Android mobile devices.
Development Framework	Flutter.
Programming Language	Dart.
Backend Service	Firebase.
Database	Cloud Firestore / Firebase Realtime Database.
Authentication	Firebase Authentication.
Notifications	Firebase Cloud Messaging.
Storage	Firebase Storage.
Location Service	GPS and Google Maps / Location Services.

6.2.1 Data Conversion

Data conversion is the process of transferring data from the old system to the new system. In the case of CYN, the existing system is mostly manual. Customers usually contact workers through phone calls, references, or local market visits. Therefore, there may not be a large digital database to convert.

However, some initial data will need to be added to the new system before deployment.

INITIAL DATA REQUIRED

Data Type	Description
Service Categories	Categories such as electrician, carpenter, mechanic, mason, labour, loader rickshaw, painter, and plumber.
Customer Profiles	Basic customer accounts for testing and initial use.
Provider Profiles	Service provider accounts with name, phone number, service category, experience, and location.
Provider Availability	Information about whether the provider is available or unavailable.
Location Data	Customer and provider location details for nearby search.
Review Data	Reviews will be generated after users complete services.

DATA CONVERSION PROCESS

The data conversion process will include the following steps:

1. Create service categories in the database.
2. Register initial service providers manually or through the application.
3. Store provider details such as name, phone number, category, experience, and location.
4. Register test customers for pilot deployment.
5. Verify that all data is stored correctly in Firebase.
6. Test service request flow using the converted or newly entered data.

Since the system is shifting from a manual process to a digital platform, most data will be created directly inside the CYN application during user registration and service usage.

6.2.2 Training

Training is required to help customers and service providers understand how to use the CYN application. Since the application will be used by different types of users, including local workers, the training process should be simple and practical.

CUSTOMER TRAINING

Customers should be trained on how to use the application for hiring service providers.

Customer training will include:

Training Area	Description
Account Creation	How to create a new customer account.
Login	How to log in using registered credentials.
Category Selection	How to select a required service category.
Location Permission	How to allow location access for nearby search.
Sending Request	How to send a service request to nearby providers.
Chat and Call	How to communicate with the provider through chat, voice call, or video call.
Tracking	How to view provider arrival status.

Review Submission How to submit rating and feedback after service completion.

SERVICE PROVIDER TRAINING

Service providers should be trained on how to receive and manage service requests.

Provider training will include:

Training Area	Description
Provider Registration	How to create a provider account.
Profile Setup	How to add service category, experience, phone number, and location.
Availability Status	How to mark status as available or unavailable.
Receiving Requests	How to view incoming service requests.
Accepting Requests	How to accept a customer request.
Rejecting Requests	How to reject a request if unavailable.
Chat and Call	How to communicate with customers.
Job Completion	How to mark a job as completed.
Rating Awareness	How reviews and ratings improve provider profile reputation.

TRAINING METHOD

Training can be provided through:

- Short user guide
- Demo video
- In-app instructions
- Practical demonstration
- Help section inside the application

The training should focus on simple steps so that both technical and non-technical users can easily understand the system.

6.3 Post Deployment Testing

Post deployment testing is performed after the application is deployed in the real environment. The purpose of this testing is to verify that the system works correctly for actual users and performs properly on real devices.

Post deployment testing helps identify issues that may not appear during development or internal testing. It ensures that the CYN application is ready for regular use.

POST DEPLOYMENT TESTING AREAS

Testing Area	Description
User Registration	Check whether customers and providers can create accounts successfully.
Login	Verify that registered users can log in without errors.
Location Access	Check whether the application can access user location correctly.
Provider Search	Verify that nearby available providers are searched correctly.
Request Delivery	Check whether service requests reach the correct provider.
Accept/Reject Request	Verify that providers can accept or reject requests.
Request Forwarding	Check whether rejected requests are forwarded to the next nearest provider.
Chat Module	Verify that messages are sent and received correctly.
Voice/Video Call	Check whether call options open and work properly.
Tracking	Verify provider arrival status and location display.
Notifications	Check whether users receive request and message notifications.
Job Completion	Verify that completed jobs are updated correctly.
Review System	Check whether reviews and ratings are saved successfully.
Profile Management	Verify that users can update profile information.

POST DEPLOYMENT TESTING RESULT

After deployment, the application should be tested with real customers and service providers. If any issue occurs, it should be recorded, analyzed, and fixed. The testing results will help improve the application's stability, performance, and user experience.

6.4 Challenges

During system conversion and deployment, several challenges may occur. These challenges must be identified and handled properly to ensure successful implementation of the CYN application.

MAJOR CHALLENGES

Challenge	Description

Location Accuracy	GPS may not always provide exact location due to weak signals or device limitations.
Internet Dependency	The application requires an active internet connection for Firebase, chat, notifications, and tracking.
User Adoption	Customers and workers may take time to shift from manual methods to the digital system.
Provider Availability	Service providers may forget to update their availability status.
Notification Delay	Push notifications may be delayed due to poor internet connection or device restrictions.
Device Compatibility	The application must work properly on different Android devices and screen sizes.
Data Security	User data such as phone numbers, location, and chat must be protected.
Call Integration	Voice and video call features may require third-party SDK integration and proper permissions.
Training Difficulty	Some local workers may need basic guidance to use the application correctly.
Trust Building	New users may need time to trust the application and service providers.

SOLUTIONS TO CHALLENGES

Challenge	**Possible Solution**
Location Accuracy	Use GPS permissions properly and allow users to confirm their location manually.
Internet Dependency	Show clear error messages when internet connection is unavailable.
User Adoption	Provide simple UI, user guide, and training.
Provider Availability	Add availability reminders for providers.
Notification Delay	Use Firebase Cloud Messaging and test notification settings on devices.
Device Compatibility	Test the app on different screen sizes and Android versions.
Data Security	Apply Firebase Authentication and database security rules.
Call Integration	Use a reliable voice/video call SDK and request microphone/camera permissions.
Training Difficulty	Provide demo videos and simple instructions.

Trust Building	Use ratings, reviews, and provider profile information.

6.5 Summary

This chapter discussed the system conversion process of the Chase Your Need application. It explained how the existing manual method of finding service providers can be converted into a digital mobile-based system.

The recommended conversion method for CYN is pilot conversion because it allows the application to be tested with a limited number of users before full deployment. The chapter also described the deployment process, data conversion, user training, post deployment testing, and challenges.

The deployment of CYN involves configuring Firebase services, building the mobile application, testing major modules, launching the system for selected users, collecting feedback, and improving the application. Since the previous system is mostly manual, most data will be created directly through user registration and service usage.

Overall, system conversion ensures that customers and service providers can smoothly shift from manual service hiring to the CYN digital platform.



Chapter 7: Conclusion

7. Introduction

This chapter presents the conclusion of the Chase Your Need (CYN) application. It includes the evaluation of the project, traceability matrix, final conclusion, and future enhancements.

The CYN application is designed to solve the problem of finding nearby local service providers through a mobile-based digital platform. The application helps customers hire workers such as electricians, carpenters, auto mechanics, masons, labor workers, loader rickshaw drivers, painters, plumbers, and other related service providers.

The main purpose of the application is to reduce the time and effort required to find trusted and available workers. Customers can search for nearby service providers using their location, send service requests, communicate through chat, voice call, or video call, track provider arrival, and submit reviews after service completion.

The system is planned to be developed using Flutter for mobile application development and Firebase for backend services such as authentication, database, notifications, storage, and real-time communication.

7.1 Evaluation

The Chase Your Need application provides a useful and practical solution for both customers and service providers. In the traditional manual system, customers usually depend on personal contacts, references, or local markets to find workers. This process is slow and does not always guarantee availability or trust. The proposed system improves this process by providing a digital platform where customers can quickly search and contact nearby available service providers.

The application was evaluated according to its main objectives and functional requirements. The system successfully supports user registration, login, role selection, service category selection, location-based search, service request handling, request acceptance or rejection, request forwarding, chat, voice/video call, tracking, job completion, and review submission.

EVALUATION POINTS

Evaluation Area	**Description**
User Registration	Customers and service providers can create accounts.
User Login	Registered users can securely log in to the application.
Role-Based Access	Customers and service providers are redirected to their relevant dashboards.
Service Category Selection	Customers can select required service categories.
Location-Based Search	The system can search nearby available providers using customer location.
Request Management	Customers can send requests, and providers can accept or reject them.

Request Forwarding	If a provider rejects a request, it can be forwarded to the next nearest provider.
Communication	Chat, voice call, and video call options improve communication between users.
Tracking	Customers can view provider arrival status.
Review and Rating	Customers can submit feedback after service completion.
Provider Profile	Reviews and ratings help improve provider profile reputation.
Usability	The application interface is simple and easy to use.

Overall, the evaluation shows that the CYN application meets its main objectives and provides an effective solution for local service hiring.

7.2 Traceability Matrix

A traceability matrix is used to map functional requirements with related test cases. It ensures that each requirement of the system has been tested properly.

Requirement ID	**Requirement Description**	**Related Test Case ID**
FR-01	User Registration	UT-03, UT-04, FT-01, FT-02
FR-02	User Login	UT-01, UT-02, FT-03, IT-01
FR-03	Role Selection	UT-05, UT-06, IT-01
FR-04	Service Category Selection	UT-07, FT-04, IT-03
FR-05	Location Access	UT-08, IT-03, IT-04
FR-06	Search Nearby Providers	FT-05, IT-04, PT-03
FR-07	Send Service Request	UT-09, FT-06, IT-05
FR-08	Accept Request	FT-07, IT-06
FR-09	Reject Request	FT-08, IT-07
FR-10	Forward Request	FT-08, IT-07
FR-11	Chat Module	UT-10, FT-09, IT-08, IT-09
FR-12	Voice and Video Call	FT-10, FT-11
FR-13	Provider Tracking	FT-12, PT-07
FR-14	Job Completion	FT-13, IT-10
FR-15	Review and Rating	UT-11, FT-14, IT-10, PT-10
FR-16	Provider Dashboard	FT-02, FT-16
FR-17	Profile Management	FT-16, IT-02

FR-18	Notifications	FT-15, IT-05, IT-12, PT-06
FR-19	Provider Availability	UT-12, IT-11

Traceability Matrix Description

The traceability matrix shows that all major requirements of the CYN application are linked with relevant test cases. This ensures that the application features have been tested through unit testing, functional testing, integration testing, and performance testing.

The matrix also helps verify that no important requirement is missed during testing. For example, login is tested through unit testing, functional testing, and integration testing. Similarly, service request handling is tested through request creation, notification delivery, provider response, and request forwarding.

7.3 Conclusion

Chase Your Need is a mobile application designed to connect customers with nearby local service providers. The application provides a digital solution for hiring workers such as electricians, carpenters, auto mechanics, masons, labor workers, loader rickshaw drivers, painters, plumbers, and other related service providers.

The application solves the common problem of finding trusted and available workers. Instead of depending on personal contacts or visiting local markets, customers can use the CYN application to search for nearby providers based on service category and location. The system sends a request to the nearest available provider. If the provider accepts the request, the customer and provider can communicate through chat, voice call, or video call. If the provider rejects the request, the system forwards the request to the next nearest available provider.

The review and rating system helps build trust between customers and service providers. It also improves the profile reputation of good workers and helps future customers choose better service providers.

The use of Flutter and Firebase makes the application efficient, real-time, scalable, and suitable for mobile users. Flutter provides a modern and smooth mobile interface, while Firebase supports authentication, database management, notifications, storage, and real-time communication.

In conclusion, the CYN application is a useful, reliable, and user-friendly platform for local service hiring. It benefits customers by saving time and helps service providers by increasing their work opportunities.

7.4 Future Work

Although the CYN application provides the main features required for hiring nearby service providers, several improvements can be added in the future to make the system more advanced and useful.

Future Enhancement	**Description**
Online Payment System	Customers can pay service charges directly through the application.
Admin Panel	Admin can manage users, providers, categories, complaints, and reports.
Provider Verification	CNIC or document verification can be added to improve trust and safety.
Complaint Management	Customers and providers can submit complaints in case of disputes.
AI-Based Recommendation	The system can recommend the best providers based on rating, distance, experience, and availability.
Advanced Search Filters	Customers can filter providers by rating, experience, price, and distance.
Emergency Service Mode	Urgent requests can be prioritized and sent to the fastest available provider.
In-App Wallet	Customers and providers can manage payments and balances inside the application.
Multi-Language Support	The application can support English, Urdu, and regional languages.
Service Packages	Providers can create fixed-price service packages for common tasks.
Provider Badge System	Top-rated providers can receive badges such as Verified, Expert, or Highly Rated.
Schedule Booking	Customers can book services for a future date and time.
Real-Time Route Tracking	Customers can view the provider's live route on the map.
Offer and Discount System	Providers can offer discounts to attract more customers.
Web-Based Admin Dashboard	Admin can monitor system activity through a web dashboard.

FINAL REMARKS

The future enhancements will make the CYN application more powerful, secure, and professional. Features such as online payment, provider verification, admin panel, complaint management, and emergency mode can increase user trust and improve the overall quality of the system.